Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

The purpose of this lab assignment is to conduct a forensic investigation of an iPhone SE running iOS 13.4.1. The investigation focuses on examining messages, contacts, call history, browsing activities, photographs and location records. It also involves identifying associated accounts and device settings. The aim is to recover and analyse relevant digital evidence while maintaining its integrity. The exercise helps develop practical skills in mobile device forensics and evidence examination.

2. Lab Report Structure

The lab report is structured around eight major investigation areas, covering evidence preservation, device accounts and settings, messages and calls, Safari browsing history and calendar records. It also includes photographs and media analysis, KnowledgeC activity, health and location evidence, and timeline reconstruction. Each section documents the examination procedures, commands, tools, evidence collected and findings. The report also includes screenshots, database queries, SHA-256 hashes and supporting evidence.

2.1. Title Page

• Lab Title: Case Study 2: iPhone SE iOS 13.4.1

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• Student ID: 2025/FWSD/11320.

• Course Name: SBT-DF205 Mobile and IoT Forensics Case Study

• Instructor Name: Aminu Idris

• Date of Submission: 4/11/2026

• Version:

2.2. Executive Summary

This lab assignment involves the forensic examination of a supplied iPhone SE iOS 13.4.1 evidence package provided by Digital Corpora. The investigation examines device information,

user accounts, messages, contacts, calls, browsing history, photographs and application activities. It also analyses health and location records to reconstruct relevant device activities. Various forensic tools and SQLite databases are used to extract and examine digital evidence. The findings are used to develop an activity timeline and establish possible connections between the recovered records.

2.3. Lab Objectives

The main objectives of this lab assignment are to:

1. Preserve the original evidence and verify its integrity using SHA-256 hashes.
2. Identify the iPhone model, iOS version, build number and associated accounts.
3. Extract and analyse messages, attachments, contacts and call history.
4. Examine Safari browsing history and available calendar records.
5. Recover and analyse photographs, media files and their metadata.
6. Investigate KnowledgeC records to understand application and device activities.
7. Examine available health and location records.
8. Reconstruct a timeline of events and draw conclusions based on the recovered digital evidence.

2.4. Tools and Resources Used

Tools/Resources	**Purpose**
Kali Linux	Main forensic examination environment.
Terminal	Executing forensic commands and managing evidence files.
SQLite3	Accessing and analysing iPhone databases.
DB Browser for SQLite	Graphical examination of SQLite databases.
ExifTool	Extracting photograph metadata, including timestamps and GPS information.
SHA-256	Verifying the integrity of digital evidence.
Python	Supporting timestamp conversion and data analysis.
Plist utilities	Examining iOS configuration and property list files.

Tools/Resources	Purpose
LibreOffice Calc	Organising evidence registers and investigation timelines.
Digital Corpora iOS 13.4.1 Dataset	Supplied iPhone SE forensic evidence.
Supplied iOS Forensics Slides	Reference materials for the investigation procedures.

2.5. Methodology

In this training case, Harbour Operations Ltd requests a forensic review of a company-issued iPhone SE following a dispute about messages, shared media and the device's movements. As the examiner, establish the associated accounts and reconstruct relevant activity. Assess whether communications, browsing, photographs and device records support a consistent account of events.

Use the original identities and historical dates in the supplied dataset. Base your conclusions on the records examined and explain any findings that remain uncertain.

Evidence and tools

Use Joshua Hickman's iOS 13.4.1 evidence package hosted by Digital Corpora. Download ios_13_4_1.zip, extract it, then extract the file system archive at Extraction/13-4-1.tar inside the package. Use that file system for the graded tasks.

Download the iOS 13.4.1 evidence package

The download is approximately 9.63 GB. Allow at least 60 GB of free disk space. Work in a Linux VM or macOS to preserve Unix paths and links. Use an archive extractor, DB Browser for SQLite or SQLite, a binary/XML plist reader, ExifTool and a spreadsheet editor. Refer to the supplied iOS slides for examination procedures.

Task 1 Evidence preservation

1. Create a handling log using case reference SBT-DF205-C2. Record the evidence source, archive names, sizes, receipt time, examiner and tool versions.

2. Calculate SHA-256 values for the source ZIP and TAR before examination and recheck them at completion. Compare with a trusted manifest for the exact files if supplied; otherwise record your values as baselines.

3. Maintain an evidence register for selected files, including evidence ID, source path, size and SHA-256. Record extraction errors, processing actions and database sidecars.

Creating a case study Folder

```
┌──(water㉿kalipurple)-[~]
└─$ mkdir -p ~/SBT-DF205-Case2/{evidence,working,exported,reports,screenshots,scripts}

┌──(water㉿kalipurple)-[~]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Projects  Public  SBT-DF205-Case1  SBT-DF205-Case2  Templates  Videos

┌──(water㉿kalipurple)-[~]
└─$ cd SBT-DF205-Case2

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2]
└─$ ls
evidence  exported  reports  screenshots  scripts  working

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2]
└─$
```

Downloading IOS 13 Image

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$ ls
ios_13_4_1.zip

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$ ls ios_13_4_1.zip -l
-rw-rw-r-- 1 water water 9629286492 Nov 21  2020 ios_13_4_1.zip

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$
```

Verifying Hashes

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$ hashdeep ios_13_4_1.zip
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/water/SBT-DF205-Case2/evidence
## $ hashdeep ios_13_4_1.zip
##
9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551,/home/water/SBT-DF205-Case2/evidence/ios_13_4_1.zip

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$
```

9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54c
cb83044ff123350890d408c154551,/home/water/SBT-DF205-Case2/evidence/ios_13_4_1.zip

Unzip to get ios 13 Extraction

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$ unzip -q ios_13_4_1.zip

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$ ls -l
total 9403612
drwxr-xr-x 6 water water       4096 Apr 19  2020 'iOS 13.4.1 Extraction'
-rw-rw-r-- 1 water water 9629286492 Nov 21  2020  ios_13_4_1.zip

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence]
└─$
```

Root level view

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence/iOS 13.4.1 Extraction/Extraction]
└─$ ls
13-4-1.tar

┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence/iOS 13.4.1 Extraction/Extraction]
└─$ tar -xvf 13-4-1.tar
Volumes/JOSH/NoTar-13-4-1/
Volumes/JOSH/NoTar-13-4-1/.ba/
Volumes/JOSH/NoTar-13-4-1/.file
Volumes/JOSH/NoTar-13-4-1/.mb/
Volumes/JOSH/NoTar-13-4-1/.Trashes
Volumes/JOSH/NoTar-13-4-1/Applications/
Volumes/JOSH/NoTar-13-4-1/bin/
Volumes/JOSH/NoTar-13-4-1/cores/
Volumes/JOSH/NoTar-13-4-1/dev/
Volumes/JOSH/NoTar-13-4-1/Developer/
Volumes/JOSH/NoTar-13-4-1/etc
Volumes/JOSH/NoTar-13-4-1/Library/
Volumes/JOSH/NoTar-13-4-1/private/
Volumes/JOSH/NoTar-13-4-1/sbin/
Volumes/JOSH/NoTar-13-4-1/System/
Volumes/JOSH/NoTar-13-4-1/tmp
Volumes/JOSH/NoTar-13-4-1/usr/
Volumes/JOSH/NoTar-13-4-1/var
Volumes/JOSH/NoTar-13-4-1/usr/bin/
Volumes/JOSH/NoTar-13-4-1/usr/include/
Volumes/JOSH/NoTar-13-4-1/usr/lib/
Volumes/JOSH/NoTar-13-4-1/usr/libexec/
Volumes/JOSH/NoTar-13-4-1/usr/local/
Volumes/JOSH/NoTar-13-4-1/usr/sbin/
Volumes/JOSH/NoTar-13-4-1/usr/share/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/Mav10.Release.bbfw
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/ramdisk/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/ramdisk/arm64SURamDisk.dmg
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/applelogo@2x~iphone.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterycharging0@2x~iphone.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterycharging1@2x~iphone.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batteryfull@2x~iphone.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterylow0@2x~iphone.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterylow1@2x~iphone.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/glyphplugin@1136~iphone-lightning.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/recoverymode@1136~iphone-lightning.im4p
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/devicetree.img4
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/FUD/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/sep-firmware.img4
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/SLAM/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/SmartIOFirmwareT7000.bin
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/SLAM/SLAM.sefw
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/fw/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/PN549_FW_02_01_5A_rev88207.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/PN549_FW_02_01_AD_37809.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/PN549_FW_03_01_5A_rev88041.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/PN549_FW_03_01_E0_53757.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/PN552_FW_02_01_4B_rev88159.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/PN552_FW_02_01_AF_59162.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/SN100V_FW_A3_01_01_3D_rev88170.plist
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/nfrestore/firmware/rf/SN100V_FW_A3_01_01_C3_rev78599.plist
```

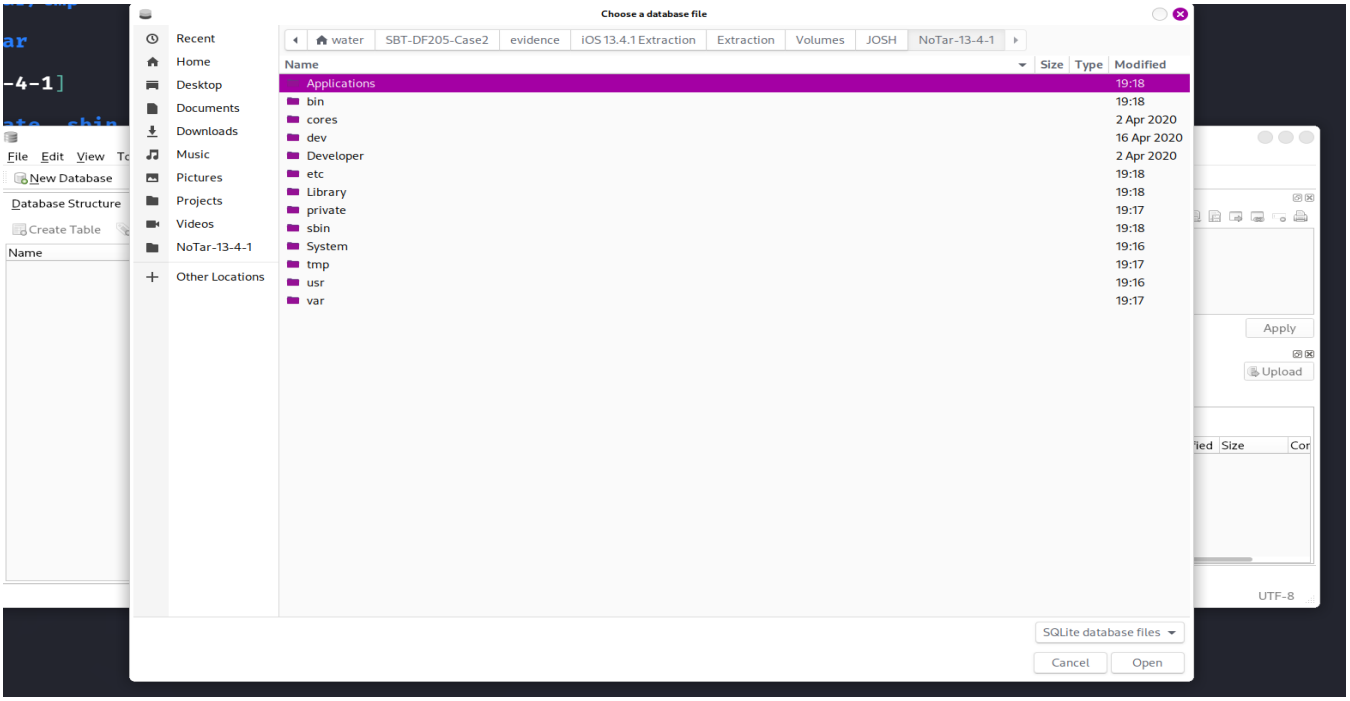
```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case2/evidence/iOS 13.4.1 Extraction/Extraction]
└─$ cd Volumes/JOSH/NoTar-13-4-1/

┌──(water㉿kalipurple)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ ls -l
total 40
drwxrwxr-x 87 water water 4096 Nov  4 19:18 Applications
drwxr-xr-x  2 water water 4096 Nov  4 19:18 bin
drwxrwxr-x  2 water water 4096 Apr  2  2020 cores
dr-xr-xr-x  2 water water 4096 Apr 16  2020 dev
drwxrwxr-x  2 water water 4096 Apr  2  2020 Developer
lrwxrwxrwx  1 water water   11 Apr  2  2020 etc → private/etc
drwxrwxr-x 17 water water 4096 Nov  4 19:18 Library
drwxr-xr-x  6 water water 4096 Nov  4 19:17 private
drwxr-xr-x  2 water water 4096 Nov  4 19:18 sbin
drwxr-xr-x  3 water water 4096 Nov  4 19:16 System
lrwxrwxrwx  1 water water   15 Apr  2  2020 tmp → private/var/tmp
drwxr-xr-x 10 water water 4096 Nov  4 19:16 usr
lrwxrwxrwx  1 water water   11 Apr  2  2020 var → private/var

┌──(water㉿kalipurple)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ ls
Applications  bin  cores  dev  Developer  etc  Library  private  sbin  System  tmp  usr  var

┌──(water㉿kalipurple)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$
```



Task 2 Device accounts and settings

1. Establish the device model, iOS version and build from configuration or log artefacts. Cite the exact source and property keys.

2. Examine Accounts3.sqlite and relevant plists to identify account associations. Corroborate one association using a second source.

3. Report two supported settings, such as location services, backup information or effective user settings. Distinguish stored configuration from proof of user activity.

```
┌──(water㉿kalipurple)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ ls
Applications  bin  cores  dev  Developer  etc  Library  private  sbin  System  tmp  usr  var

┌──(water㉿kalipurple)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ cd var

┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ ls
audit     buddy           containers             empty      installd           Keychains  lock  'Managed Preferences'  MobileDevice           networkd      run      staged_system_apps  wireless
backups   cache           db                     folders    iomfb_bics_daemon  lib        log   mobile                 MobileSoftwareUpdate   preferences   select   tmp
binpack   checkra1n.dmg   dropbear_rsa_host_key  hardware   keybags            local      logs  MobileAsset            msgs                   root          spool    vm

┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ cd installd

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/installd]
└─$ ls
Library

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/installd]
└─$ cd Library

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/installd/Library]
└─$ ls
Caches  Logs  MobileInstallation  Preferences

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/installd/Library]
└─$ cd MobileInstallation

┌──(water㉿kalipurple)-[~/…/var/installd/Library/MobileInstallation]
└─$ ls
BackedUpState  DiskImagesInfo.plist  LastBuildInfo.plist  MigrationInfo.plist  RoleUserMigration.plist
```

The Software version Information

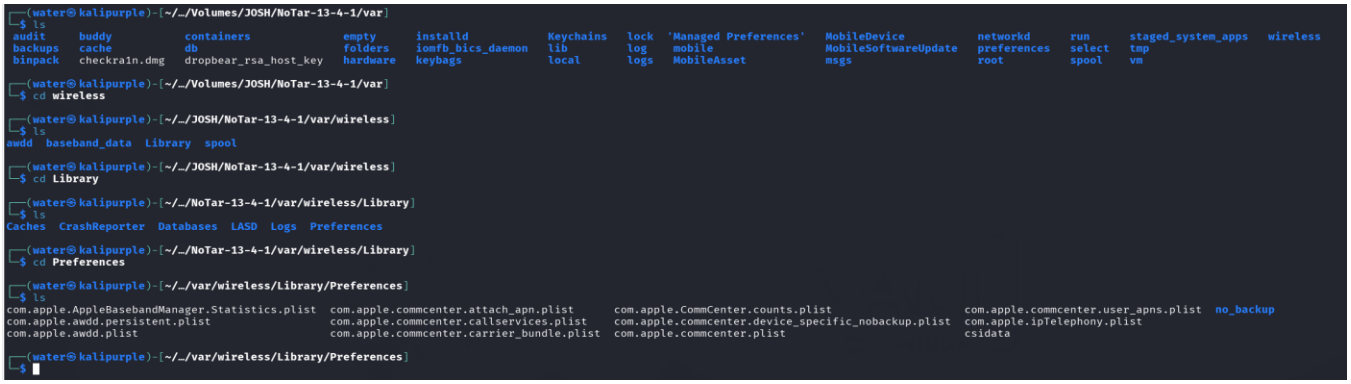


```
Session  Actions  Edit  View  Help

┌──(water㉿kalipurple)-[~/…/var/installd/Library/MobileInstallation]
└─$ plistutil -i LastBuildInfo.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>SystemImageID</key>
        <string>747CABD1-394F-4F90-92E9-CC6B5E1699DE</string>
        <key>Version</key>
        <string>Version</string>
        <key>ProductVersion</key>
        <string>13.4.1</string>
        <key>BuildID</key>
        <string>9695BE08-7547-11EA-B2DF-F0563A0B6C5E</string>
        <key>ProductBuildVersion</key>
        <string>17E262</string>
        <key>ProductCopyright</key>
        <string>1983-2020 Apple Inc.</string>
        <key>Build</key>
        <string>Build</string>
        <key>ProductName</key>
        <string>iPhone OS</string>
        <key>FullVersionString</key>
        <string>Version 13.4.1 (Build 17E262)</string>
</dict>
</plist>

┌──(water㉿kalipurple)-[~/…/var/installd/Library/MobileInstallation]
└─$
```

Confirming Phone number



```
┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ ls
 audit     buddy          containers             empty      installd           Keychains   lock   'Managed Preferences'   MobileDevice           networkd      run      staged_system_apps   wireless
 backups   cache          db                     folders    iomfb_bics_daemon  lib         log    mobile                  MobileSoftwareUpdate   preferences   select   tmp
 binpack   checkra1n.dmg  dropbear_rsa_host_key  hardware   keybags            local       logs   MobileAsset             msgs                   root          spool    vm

┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ cd wireless

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/wireless]
└─$ ls
awdd  baseband_data  Library  spool

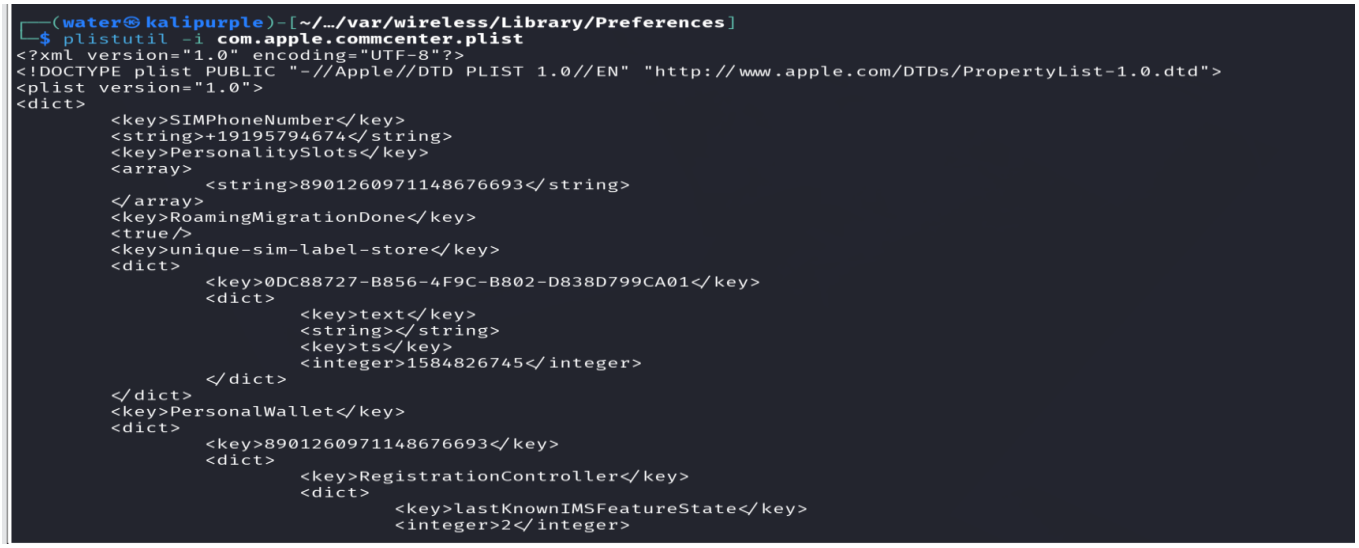
┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/wireless]
└─$ cd Library

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/wireless/Library]
└─$ ls
Caches  CrashReporter  Databases  LASD  Logs  Preferences

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/wireless/Library]
└─$ cd Preferences

┌──(water㉿kalipurple)-[~/…/var/wireless/Library/Preferences]
└─$ ls
com.apple.AppleBasebandManager.Statistics.plist  com.apple.commcenter.attach_apn.plist      com.apple.CommCenter.counts.plist                  com.apple.commcenter.user_apns.plist  no_backup
com.apple.awdd.persistent.plist                  com.apple.commcenter.callservices.plist   com.apple.commcenter.device_specific_nobackup.plist  com.apple.ipTelephony.plist
com.apple.awdd.plist                             com.apple.commcenter.carrier_bundle.plist  com.apple.commcenter.plist                          csidata

┌──(water㉿kalipurple)-[~/…/var/wireless/Library/Preferences]
└─$
```



```
┌──(water㉿kalipurple)-[~/…/var/wireless/Library/Preferences]
└─$ plistutil -i com.apple.commcenter.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>SIMPhoneNumber</key>
        <string>+19195794674</string>
        <key>PersonalitySlots</key>
        <array>
                <string>8901260971148676693</string>
        </array>
        <key>RoamingMigrationDone</key>
        <true/>
        <key>unique-sim-label-store</key>
        <dict>
                <key>0DC88727-B856-4F9C-B802-D838D799CA01</key>
                <dict>
                        <key>text</key>
                        <string></string>
                        <key>ts</key>
                        <integer>1584826745</integer>
                </dict>
        </dict>
        <key>PersonalWallet</key>
        <dict>
                <key>8901260971148676693</key>
                <dict>
                        <key>RegistrationController</key>
                        <dict>
                                <key>lastKnownIMSFeatureState</key>
                                <integer>2</integer>
```

The device ICCID

```
┌──(water㉿kalipurple)-[~/…/var/wireless/Library/Preferences]
└─$ plistutil -i com.apple.commcenter.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
```

```
        <true/>
        <key>LastKnownICCID</key>
        <string>8901260971148676693</string>
        <key>EnhancedLQMLinkPowerCostRegistration</key>
        <false/>
        <key>NetworkPhoneNumberICCID</key>
        <string>8901260971148676693</string>
        <key>LastKnownServingMcc</key>
        <integer>310</integer>
        <key>NetworkPhoneNumber</key>
        <string>+19195794674</string>
```

The device IMS

```
                <key>CarrierEntitlements</key>
                <dict>
                        <key>lastGoodImsi</key>
                        <string>310260974867669</string>
                        <key>kEntitlementsSelfRegistrationUpdateImsi</key>
                        <string>310260974867669</string>
                        <key>kEntitlementsSelfRegistrationUpdateImei</key>
                        <string>355800076093966</string>
                        <key>phone-number-prefetched</key>
                        <true/>
```

The device MNC and MCC

```
        <true/>
        <key>LastKnownServingMnc</key>
        <integer>260</integer>
        <key>PhoneNumberChangeReport</key>
        <true/>
```

```
        <key>LastKnownServingMcc</key>
        <integer>310</integer>
        <key>NetworkPhoneNumber</key>
```

The device serial number

```
┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ ls
audit    buddy          containers            empty     installd          Keychains  lock  'Managed Preferences'  MobileDevice          networkd     run     staged_system_apps  wireless
backups  cache          db                    folders   iomfb_bics_daemon  lib        log   mobile                 MobileSoftwareUpdate  preferences  select  tmp
binpack  checkra1n.dmg  dropbear_rsa_host_key  hardware  keybags           local      logs  MobileAsset            msgs                  root         spool   vm

┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ cd logs

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/logs]
└─$ ls
AKSSetup.log  AppleSupport  BurnIn  CrashReporter  keybagd.log.0  keybagd.log.1  keybagd.log.2  lockdownd.log  lockdownd.log.1  lockdown_service.log  mediaserverd  OTACrashCopier

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/logs]
└─$ cd AppleSupport

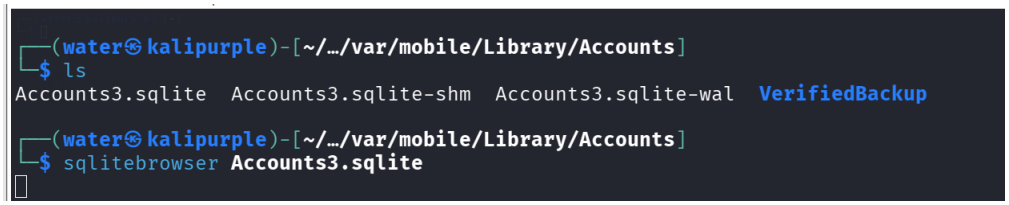
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/logs/AppleSupport]
└─$ ls
general-18356-17D50.log  general-18359-17D50.log  general-18364-17D50.log  general-18365-17E262.log  general.log

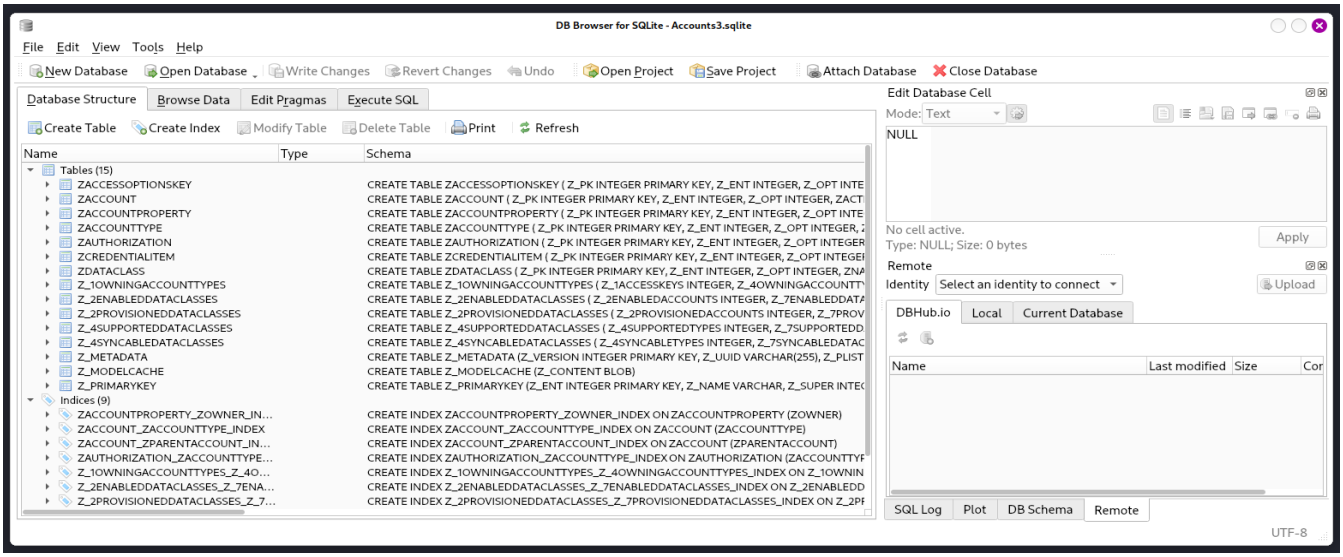
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/logs/AppleSupport]
└─$ cat general.log
{"report":"Device Software Diagnostic Log","serial":"DX3T126VH2XV","os_build":"iPhone OS 13.4.1 (17E262)","updated":"2020-04-13 15:15:32 -0400","model":"iPhone8,4","version":4,"installed":"2020-03-21 17:39:05 -0400"}

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/logs/AppleSupport]
└─$
```

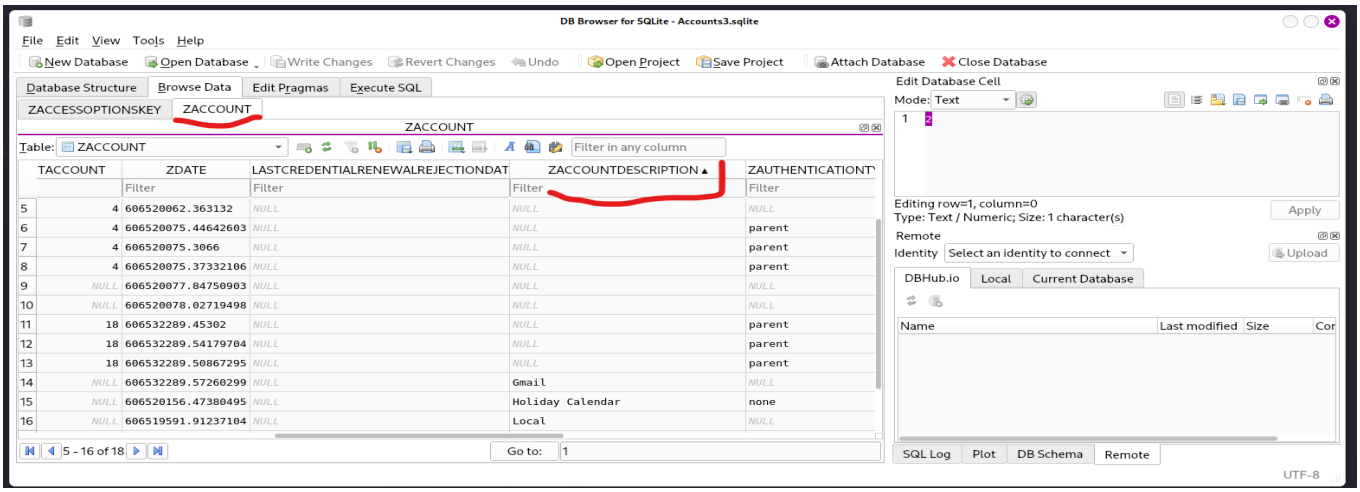
Checking Account information











BackUp Information

```
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ ls
Accessibility            'Carrier Bundle.bundle'                 Cookies                    ExternalAccessory      MedicalID                PersonalizationPortrait   Suggestions
Accounts                 'Carrier Bundles'                       CoreAccessories            FairPlay               MessagesMetaData         PowerUI                   SyncedPreferences
AddressBook              CarrierDefault.bundle                   CoreAS                     FrontBoard             MobileContainerManager   Preferences               TCC
AggregateDictionary      ClassKit                                CoreBehavior               GameKit                Mobile Documents'        Recents                   terminus
AirPlayRoutePrediction   com.apple.assistant.Contacts            CoreDuet                   Health                 MobileInstallation       RelevanceEngine           Trial
AppConduit               com.apple.companionappd                 CoreFollowUp               homed                  MusicLibrary             Reminders                 UserConfigurationProfiles
ApplePushService         com.apple.icloud.searchpartyd           coreidvd                   IdentityServices       NanoMusicSync            ReplayKit                 UserFonts
'Application Support'    com.apple.internal.ck                   CrashReporter              Inboxes                NanoPhotos               ResponseKit               UserNotifications
Assistant                com.apple.iTunesCloud                   Cydia                      Intents                NanoPreferencesSync      Safari                    UserNotificationsServer
Avatar                   com.apple.itunesstored                  DataAccess                 Keyboard               NanoTimeKit              SafariSafeBrowsing        Voicemail
BackBoard                com.apple.nsurlsessiond                 DES                        KeyboardServices       News                     'Saved Application State' VoiceServices
BulletinBoard            com.apple.nsurlsessiond-launchd         DeviceRegistry             LASD                   NFStorage                SecureElementService      VoiceTrigger
Caches                   com.apple.sensorkitd                    DeviceRegistry.dontBackUp  Library                Notes                    Sharing                   WatchConnectivity
Calendar                 com.apple.siri.inference                DeviceRegistry.state       locationd              OnDemandResources        Shortcuts                 WebClips
CallDirectory            com.apple.VideoSubscriberAccount.DeveloperService  DifferentialPrivacy    Logs                   Operator1Bundle.bundle   SMS                       WebKit
CallHistoryDB            com.apple.WatchListKit                  dmd                        mad                    'Operator Bundle.bundle' SoftwareUpdate            WelcomeMat
CallHistoryTransactions  Contacts                                DoNotDisturb               Mail                   OSAnalytics              SplashBoard
CallServices             ContactsMetadata                        DuetActivityScheduler      MediaRemote            OTALogging               Spotlight
CarrierBundle.bundle     ControlCenter                           DuetExpertCenter           MediaStream            Passes                   SpringBoard

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ cd Preferences

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Preferences]
└─$ ls
com.apple.Accessibility.Collection.plist                com.apple.health.shared.plist                                  com.apple.NanoRegistry.plist
com.apple.Accessibility.plist                           com.apple.HearingAids.plist                                    com.apple.NanoTimeKit.daemon.plist
com.apple.Accessibility.SwitchControl.plist             com.apple.homed.notbackedup.plist                              com.apple.NanoTimeKit.face.plist
com.apple.accessoryd.plist                              com.apple.homed.plist                                          com.apple.nanoweatherprefs.plist
com.apple.accountsd.plist                               com.apple.Home.group.plist                                     com.apple.NewDeviceOutreach.plist
com.apple.ActivitySharing.plist                         com.apple.Home.plist                                           com.apple.newscore2.plist
com.apple.AdLib.plist                                   com.apple.homesharing.plist                                    com.apple.newscore.plist
com.apple.adtracking.plist                              com.apple.Home.wallpaper.plist                                 com.apple.newsd.plist
com.apple.aggregated.addaily.plist                      com.apple.iapd.plist                                           com.apple.NTKFaceSnapshotService.plist
com.apple.aggregated.plist                              com.apple.iCloudDriveApp.plist                                 com.apple.operator_1.merged.plist
com.apple.airplay.plist                                 com.apple.icloud.findmydeviced.FMIPAccounts.notbackedup.plist  com.apple.operator_1.plist
com.apple.AirPlayRoutePrediction.plist                  com.apple.icloud.findmydeviced.FMIPAccounts.plist              com.apple.operator.merged.plist
com.apple.AppleMediaServices.notbackedup.plist          com.apple.icloud.findmydeviced.notbackedup.plist               com.apple.operator.plist
com.apple.AppleMediaServices.plist                      com.apple.icloud.findmydeviced.plist                           com.apple.osanalytics.addaily.plist
com.apple.AppPredictionWidget.defaults.plist            com.apple.icloud.findmydeviced.postwipe.plist                  com.apple.pairedsyncd.plist
com.apple.appstored.plist                               com.apple.icloud.findmydeviced.watch.notbackedup.plist         com.apple.parsecd.plist
com.apple.AppStore.plist                                com.apple.icloud.fmfd.notbackedup.plist                        com.apple.passd.plist
com.apple.AppSupport.plist                              com.apple.icloud.fmfd.plist                                    com.apple.persistentconnection-mcc.plist
com.apple.apsd.plist                                    com.apple.icloud.fmflocatord.notbackedup.plist                 com.apple.photoanalysisd.plist
com.apple.asd.plist                                     com.apple.icloud.fmflocatord.plist                             com.apple.Preferences.plist
com.apple.assetsd.plist                                 com.apple.icloud.fmip.clientconfiguration.plist                com.apple.private.health.menstrual-cycles.plist
com.apple.assistant.backedup.plist                      com.apple.icloud.fmip.voiceassistantsync.plist                 com.apple.proactive.PersonalizationPortrait.plist
com.apple.assistant.plist                               com.apple.icloud.searchpartyd.plist                            com.apple.protectedcloudstorage.protectedcloudkeysyncing.plist
com.apple.assistant.support.plist                       com.apple.identityservicesd.plist                              com.apple.PrototypeTools.plist
com.apple.AssistiveTouch.plist                          com.apple.identityservices.idstatuscache.plist                 com.apple.purplebuddy.notbackedup.plist
com.apple.atc.plist                                     com.apple.ids.deviceproperties.plist                           com.apple.purplebuddy.plist
com.apple.AvatarUI.Staryu.plist                         com.apple.ids.plist                                            com.apple.rapport.plist
com.apple.avfoundation.frecents.plist                   com.apple.ids.subservices.plist                                com.apple.remindd.babysitter.plist
com.apple.backboardd.plist                              com.apple.ImageCaptureFramework.plist                          com.apple.remindd.plist
com.apple.BatteryCenter.BatteryWidget.plist             com.apple.imagent.plist                                        com.apple.routined.plist
com.apple.bird.plist                                    com.apple.IMCoreSpotlight.plist                                com.apple.rtcreporting.plist
com.apple.bluetooth.settings.plist                      com.apple.imdbadgeutilities.plist                              com.apple.SafariBookmarksSyncAgent.plist
com.apple.Bridge.plist                                  com.apple.imdpersistence.IMDPersistenceAgent.plist             com.apple.SafariCloudHistoryPushAgent.plist
com.apple.BTServer.plist                                com.apple.imdsmsrecordstore.plist                              com.apple.Safari.SafeBrowsing.plist
com.apple.bulletinboard.plist                           com.apple.imessage.bag.plist                                   com.apple.Safari.SandboxBroker.plist
com.apple.bulletindistributor.plist                     com.apple.imessage.plist                                       com.apple.ScreenTimeAgent.plist
```

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Preferences]
└─$ plistutil -i com.apple.mobile.ldbackup.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>CloudBackupEnabled</key>
        <true/>
        <key>Version</key>
        <string>2.0</string>
        <key>LastiTunesBackupTZ</key>
        <string>EDT</string>
        <key>LastiTunesBackupDate</key>
        <integer>608744930</integer>
        <key>LastCloudBackupTZ</key>
        <string>EDT</string>
        <key>WillEncrypt</key>
        <true/>
        <key>RequiresEncryption</key>
        <integer>0</integer>
        <key>LastCloudBackupDate</key>
        <integer>608693130</integer>
</dict>
</plist>

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Preferences]
└─$
```

Location Service

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Preferences]
└─$ plistutil -i com.apple.locationd.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
        <key>LastSystemVersion</key>
        <string>iPhone OS13.4.1/17E262</string>
        <key>MotionFitnessDailyToken</key>
        <string>A13F0FBE-92A7-43CD-B4CF-4D4A92D9C148</string>
        <key>kCLVersionOfLastFullSync</key>
        <integer>4</integer>
        <key>CLSilo.Version</key>
        <integer>1</integer>
        <key>ObsoleteDataDeleted</key>
        <true/>
        <key>OscarStepCounting</key>
        <true/>
        <key>LocationServicesEnabledIn8.0</key>
        <true/>
</dict>
</plist>

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Preferences]
└─$
```

Task 3 Messages contacts and calls

1. Locate private/var/mobile/Library/SMS/sms.db and inspect its schema. Extract at least five messages and reconstruct one conversation, showing record IDs, participants, content, direction, service type and raw and converted timestamps.

2. Trace one message attachment record to its local file where available. Explain the relationship and any unavailable content. Validate status meanings against the schema and application version.

3. Use AddressBook.sqlitedb to associate at least two message or call identifiers with stored contacts where matches exist. Examine at least three calls in CallHistory.storedata, including direction, time and duration.

```
┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ ls -l
total 7008
drwx------   2 water water      4096 Apr 13   2020  audit
drwxr-xr-x   2 water water      4096 Oct 28   2006  backups
drwxr-xr-x   3 water water      4096 Nov  4 19:18  binpack
drwx------   2 water water      4096 Jan 18   2020  buddy
drwxr-xr-x   4 water water      4096 Nov  4 19:18  cache
-rw-r--r--   1 water water   7025305 Apr 12   2020  checkra1n.dmg
drwxr-xr-x   5 water water      4096 Nov  4 19:18  containers
drwxr-xr-x  25 water water      4096 Nov  4 19:18  db
-rw-------   1 water water       805 Apr 12   2020  dropbear_rsa_host_key
drwxr-xr-x   2 water water      4096 Jan 18   2020  empty
drwxr-xr-x   2 water water      4096 Jan 18   2020  folders
drwxr-xr-x   2 water water      4096 Apr 13   2020  hardware
drwxr-xr-x   3 water water      4096 Nov  4 19:18  installd
drwxr-xr-x   2 water water      4096 Mar 21   2020  iomfb_bics_daemon
drwx------   3 water water      4096 Nov  4 19:18  keybags
drwxr-xr-x   5 water water      4096 Nov  4 19:18  Keychains
drwxr-xr-x   5 water water      4096 Nov  4 19:18  lib
drwxrwxr-x   2 water water      4096 Oct 28   2006  local
drwxrwxr-x   2 water water      4096 Oct 28   2006  lock
drwxr-xr-x   9 water water      4096 Nov  4 19:18  log
drwxr-xr-x   7 water water      4096 Nov  4 19:18  logs
drwxr-xr-x   3 water water      4096 Nov  4 19:18 'Managed Preferences'
drwx--x--x   9 water water      4096 Nov  4 19:17  mobile
drwxr-xr-x   5 water water      4096 Nov  4 19:17  MobileAsset
drwxr-xr-x   3 water water      4096 Nov  4 19:17  MobileDevice
drwxr-xr-x   9 water water      4096 Nov  4 19:17  MobileSoftwareUpdate
drwxr-xr-x   2 water water      4096 Jan 18   2020  msgs
drwxr-xr-x   3 water water      4096 Nov  4 19:17  networkd
drwxr-xr-x   4 water water      4096 Nov  4 19:17  preferences
drwxr-xr-x   4 water water      4096 Nov  4 19:17  root
drwxrwxr-x   2 water water      4096 Nov  4 19:17  run
drwxr-xr-x   2 water water      4096 Apr 13   2020  select
drwxr-xr-x   2 water water      4096 Oct 28   2006  spool
drwxrwxr-x   2 water water      4096 Apr 13   2020  staged_system_apps
drwxrwxr-x  39 water water      4096 Nov  4 19:17  tmp
drwxr-xr-x   2 water water      4096 Nov  4 19:17  vm
drwxr-xr-x   6 water water      4096 Mar 21   2020  wireless

┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$
```

SMS Investigation

```
┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ cd mobile

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/mobile]
└─$ ls
Applications  Containers  Documents  Downloads  Library  Media  MobileSoftwareUpdate

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/mobile]
└─$ cd Library

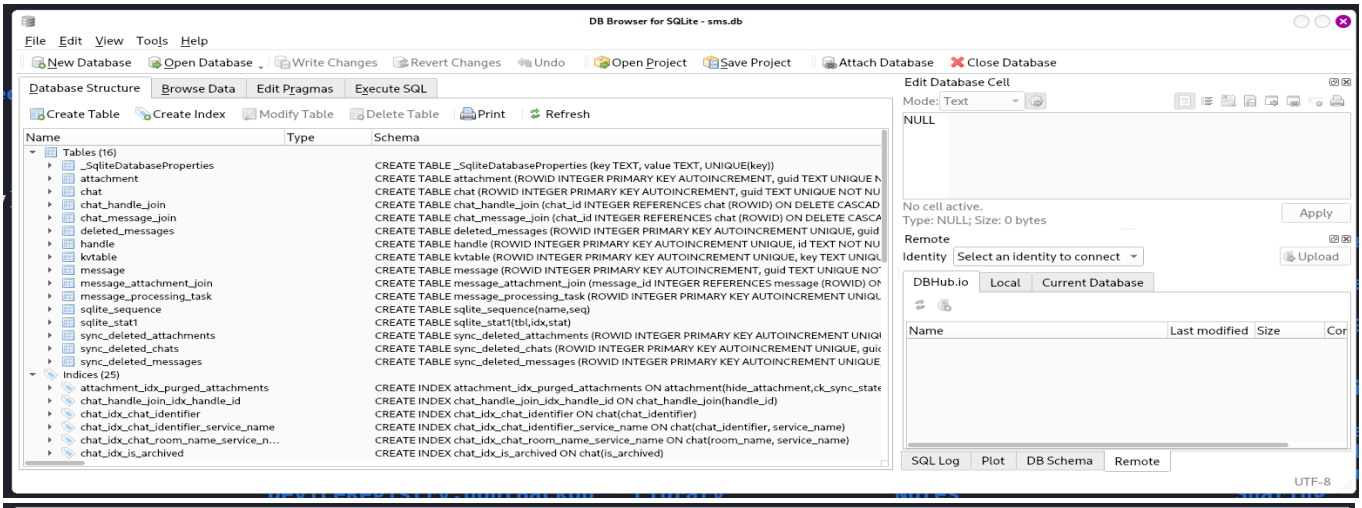
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ ls
 Accessibility             'Carrier Bundle.bundle'                        Cookies                  ExternalAccessory     MedicalID                PersonalizationPortrait   Suggestions
 Accounts                  'Carrier Bundles'                              CoreAccessories          FairPlay              MessagesMetaData         PowerUI                   SyncedPreferences
 AddressBook               CarrierDefault.bundle                          CoreAS                   FrontBoard            MobileContainerManager   Preferences               TCC
 AggregateDictionary       ClassKit                                       CoreBehavior             GameKit               'Mobile Documents'       Recents                   terminus
 AirPlayRoutePrediction    com.apple.assistant.Contacts                   CoreDuet                 Health                MobileInstallation       RelevanceEngine           Trial
 AppConduit                com.apple.companionappd                        CoreFollowUp             homed                 MusicLibrary             Reminders                 UserConfigurationProfiles
 ApplePushService          com.apple.icloud.searchpartyd                  coreidvd                 IdentityServices      NanoMusicSync            ReplayKit                 UserFonts
'Application Support'      com.apple.internal.ck                          CrashReporter            Inboxes               NanoPhotos               ResponseKit               UserNotifications
 Assistant                 com.apple.iTunesCloud                          Cydia                    Intents               NanoPreferencesSync      Safari                    UserNotificationsServer
 Avatar                    com.apple.itunesstored                         DataAccess               Keyboard              NanoTimeKit              SafariSafeBrowsing        Voicemail
 BackBoard                 com.apple.nsurlsessiond                        DES                      KeyboardServices      News                    'Saved Application State'  VoiceServices
 BulletinBoard             com.apple.nsurlsessiond-launchd                DeviceRegistry           LASD                  NFStorage                SecureElementService      VoiceTrigger
 Caches                    com.apple.sensorkitd                           DeviceRegistry.dontBackUp Library              Notes                    Sharing                   WatchConnectivity
 Calendar                  com.apple.siri.inference                       DeviceRegistry.state     locationd             OnDemandResources        Shortcuts                 WebClips
 CallDirectory             com.apple.VideoSubscriberAccount.DeveloperService DifferentialPrivacy   Logs                  Operator1Bundle.bundle   SMS                       WebKit
 CallHistoryDB             com.apple.WatchListKit                         dmd                      mad                  'Operator Bundle.bundle' SoftwareUpdate            WelcomeMat
 CallHistoryTransactions   Contacts                                       DoNotDisturb             Mail                  OSAnalytics              SplashBoard
 CallServices              ContactsMetadata                               DuetActivityScheduler    MediaRemote           OTALogging               Spotlight
 Carrier1Bundle.bundle     ControlCenter                                  DuetExpertCenter         MediaStream           Passes                   SpringBoard

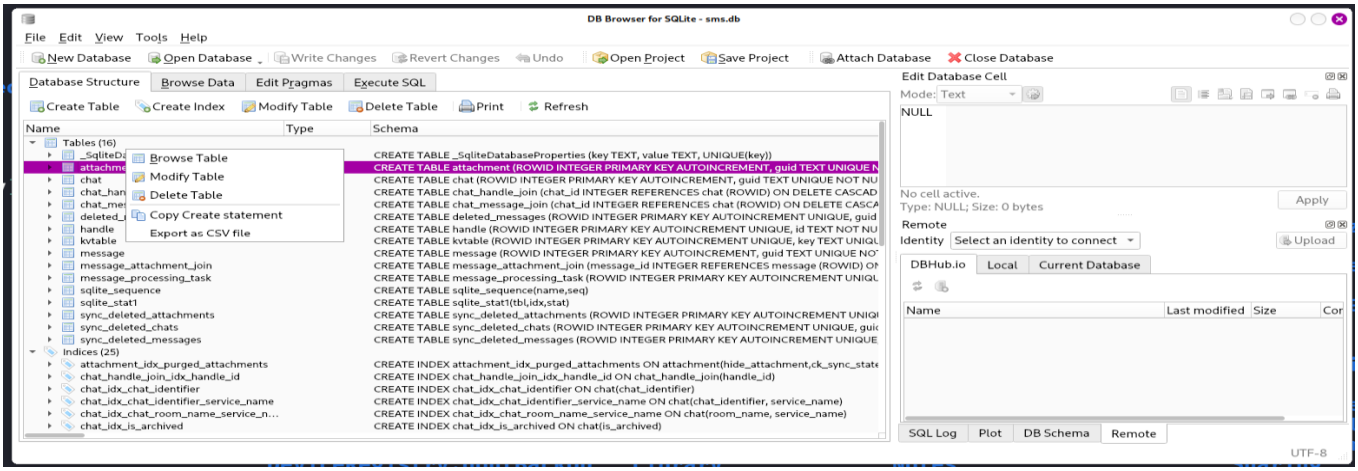
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ cd SMS
```

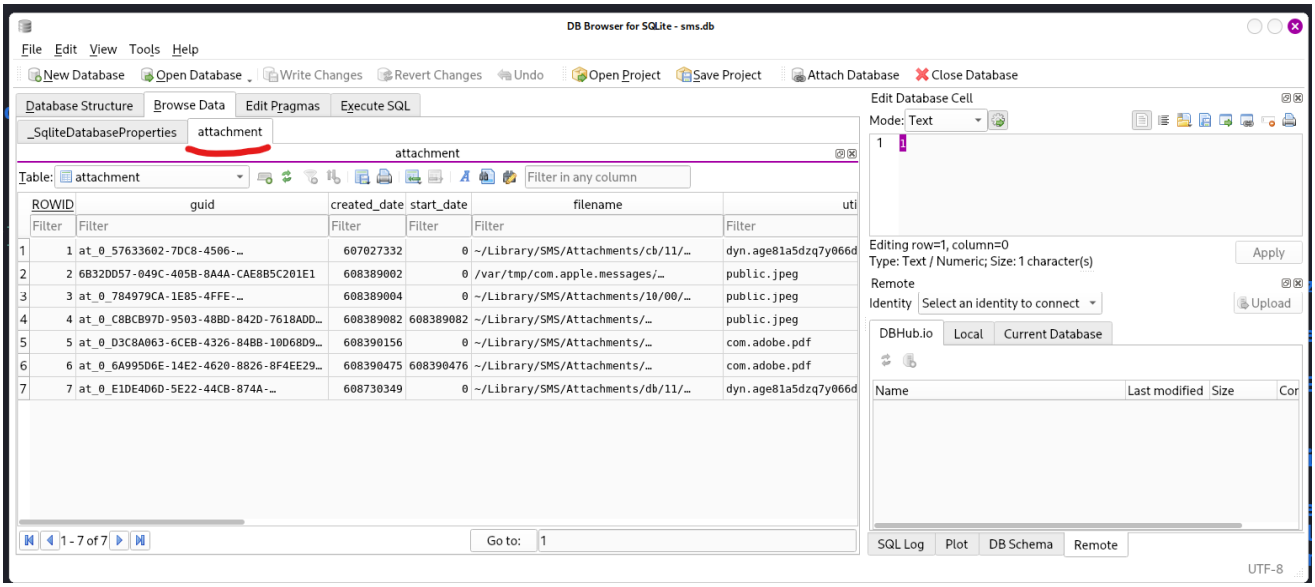
```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/SMS]
└─$ ls
Attachments  CloudKitMetaData  com.apple.messages.geometrycache_v3.plist  Drafts  PluginMetaDataCache  prewarm.db  prewarm.db-shm  prewarm.db-wal  sms.db  sms.db-shm  sms.db-wal

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/SMS]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/SMS]
└─$ sqlitebrowser sms.db
```





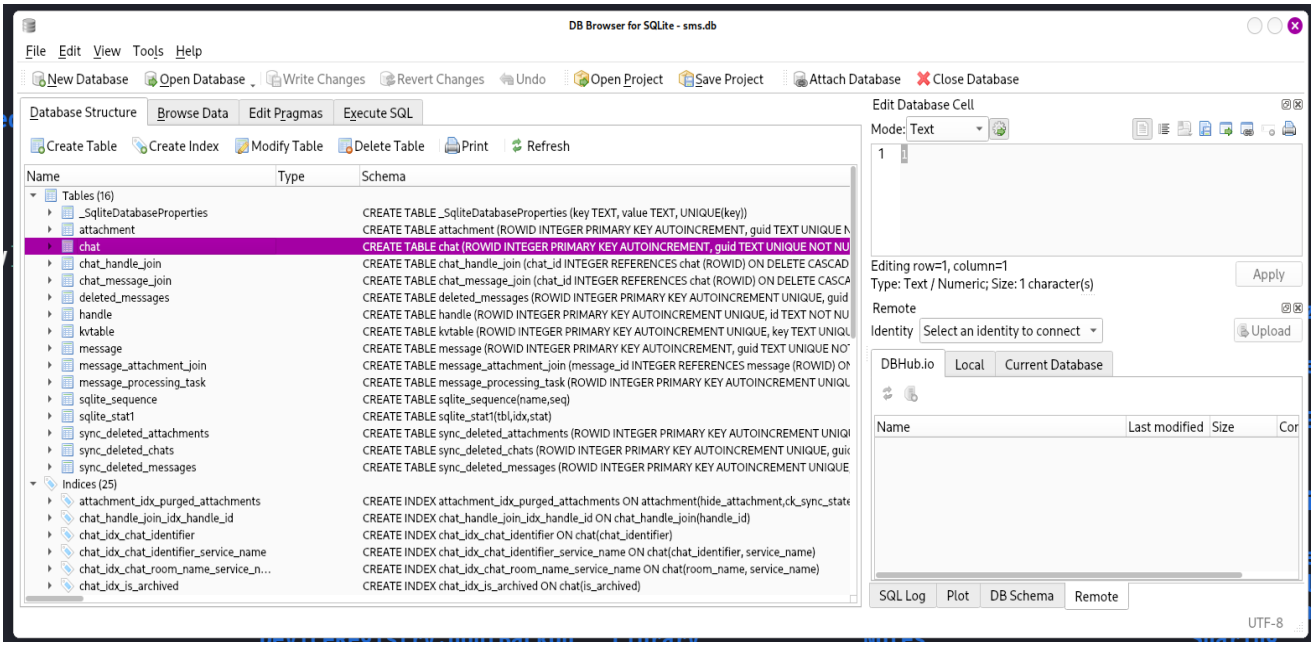


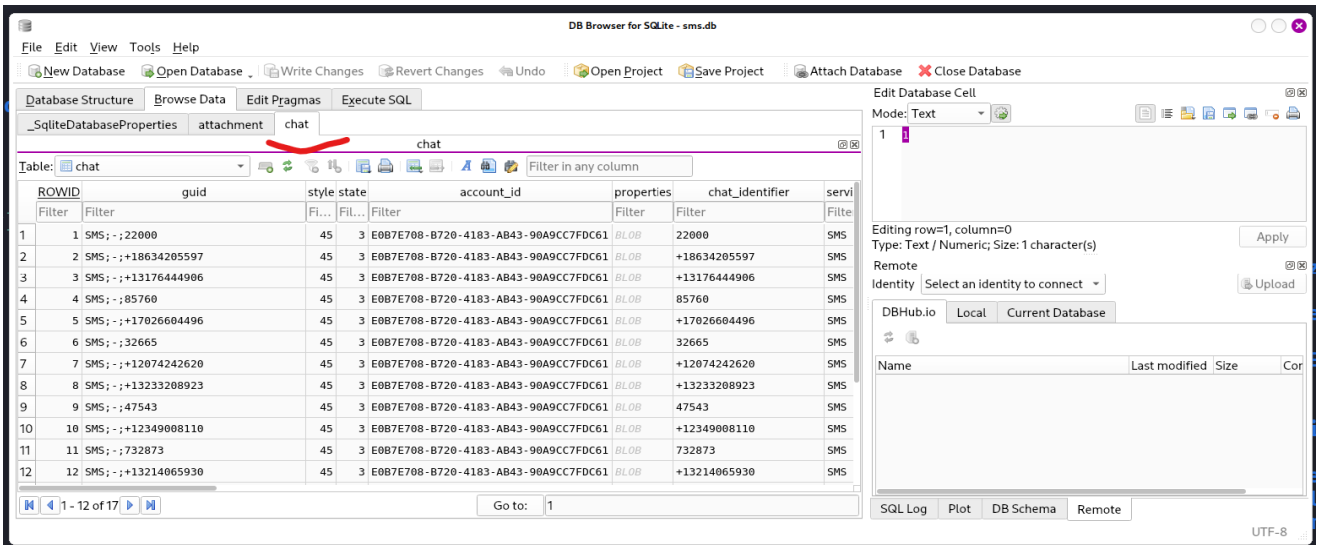

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/SMS]
└─$ echo $((978307200+607027332))
1585334532

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/SMS]
└─$ date -d @1585334532
Fri Mar 27 02:42:12 PM EDT 2020

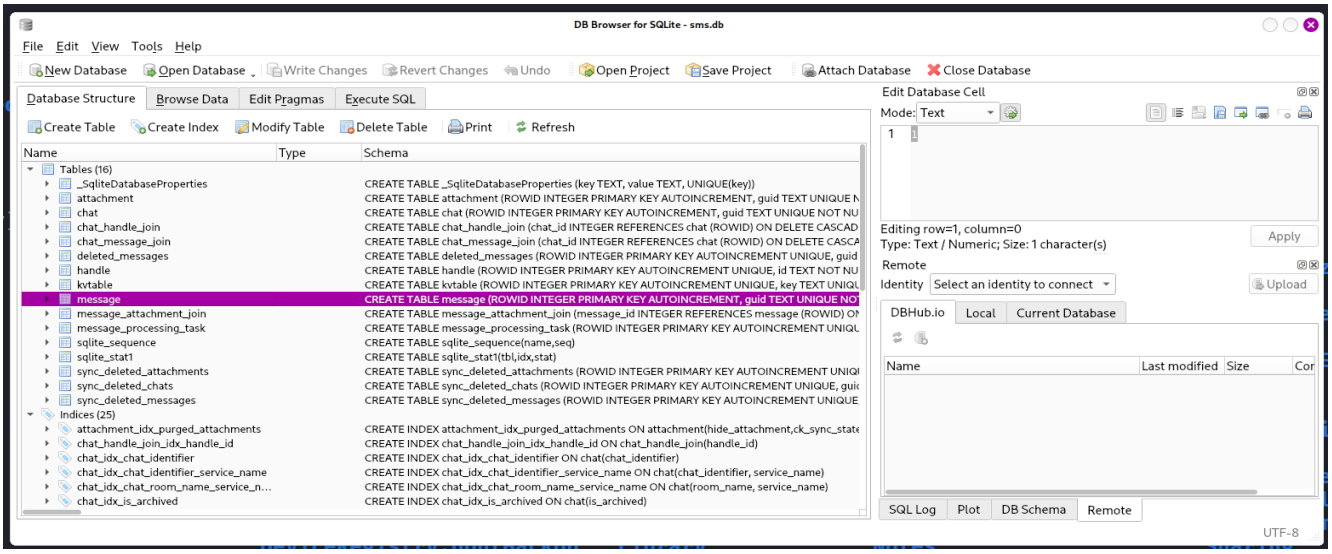
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/SMS]
└─$
```

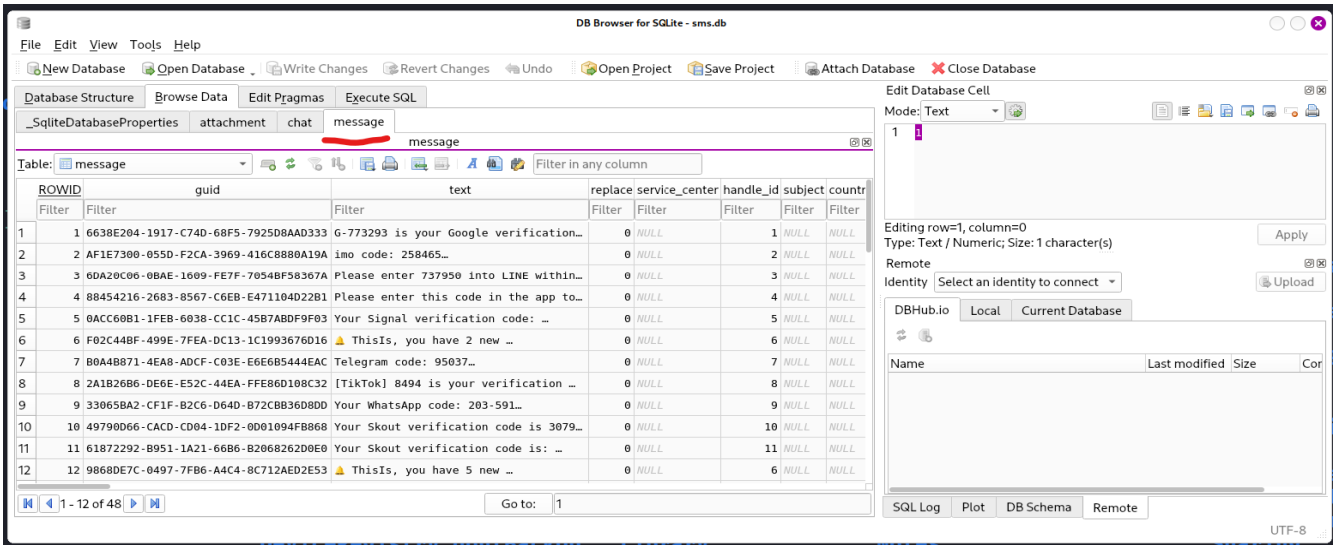
Chat




DB Browser for SQLite - sms.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
_SqliteDatabaseProperties attachment chat
chat
Table: chat
Filter in any column
ROWID guid style state account_id properties chat_identifier servi
1 1 SMS;-;22000 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB 22000 SMS
2 2 SMS;-;+18634205597 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +18634205597 SMS
3 3 SMS;-;+13176444906 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +13176444906 SMS
4 4 SMS;-;85760 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB 85760 SMS
5 5 SMS;-;+17026604496 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +17026604496 SMS
6 6 SMS;-;32665 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB 32665 SMS
7 7 SMS;-;+12074242620 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +12074242620 SMS
8 8 SMS;-;+13233208923 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +13233208923 SMS
9 9 SMS;-;47543 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB 47543 SMS
10 10 SMS;-;+12349008110 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +12349008110 SMS
11 11 SMS;-;732873 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB 732873 SMS
12 12 SMS;-;+13214065930 45 3 E0B7E708-B720-4183-AB43-90A9CC7FDC61 BLOB +13214065930 SMS
1 - 12 of 17
Go to: 1
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8

Message


DB Browser for SQLite - sms.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
Create Table Create Index Modify Table Delete Table Print Refresh
Name Type Schema
Tables (16)
_SqliteDatabaseProperties CREATE TABLE _SqliteDatabaseProperties (key TEXT, value TEXT, UNIQUE(key))
attachment CREATE TABLE attachment (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE N
chat CREATE TABLE chat (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NOT NU
chat_handle_join CREATE TABLE chat_handle_join (chat_id INTEGER REFERENCES chat (ROWID) ON DELETE CASCAD
chat_message_join CREATE TABLE chat_message_join (chat_id INTEGER REFERENCES chat (ROWID) ON DELETE CASCA
deleted_messages CREATE TABLE deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid
handle CREATE TABLE handle (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, id TEXT NOT NU
kvtable CREATE TABLE kvtable (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, key TEXT UNIQU
message CREATE TABLE message (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NO
message_attachment_join CREATE TABLE message_attachment_join (message_id INTEGER REFERENCES message (ROWID) ON
message_processing_task CREATE TABLE message_processing_task (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQU
sqlite_sequence CREATE TABLE sqlite_sequence(name,seq)
sqlite_stat1 CREATE TABLE sqlite_stat1(tbl,idx,stat)
sync_deleted_attachments CREATE TABLE sync_deleted_attachments (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQ
sync_deleted_chats CREATE TABLE sync_deleted_chats (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, gui
sync_deleted_messages CREATE TABLE sync_deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE
Indices (25)
attachment_idx_purged_attachments CREATE INDEX attachment_idx_purged_attachments ON attachment(hide_attachment,ck_sync_state
chat_handle_join_idx_handle_id CREATE INDEX chat_handle_join_idx_handle_id ON chat_handle_join(handle_id)
chat_idx_chat_identifier CREATE INDEX chat_idx_chat_identifier ON chat(chat_identifier)
chat_idx_chat_identifier_service_name CREATE INDEX chat_idx_chat_identifier_service_name ON chat(chat_identifier, service_name)
chat_idx_chat_room_name_service_n... CREATE INDEX chat_idx_chat_room_name_service_name ON chat(room_name, service_name)
chat_idx_is_archived CREATE INDEX chat_idx_is_archived ON chat(is_archived)
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8


DB Browser for SQLite - sms.db
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
_SqliteDatabaseProperties attachment chat message
message
Table: message
Filter in any column
ROWID guid text replace service_center handle_id subject countr
1 1 6638E204-1917-C74D-68F5-7925D8AAD333 G-773293 is your Google verification… 0 NULL 1 NULL NULL
2 2 AF1E7300-055D-F2CA-3969-416C8880A19A imo code: 258465… 0 NULL 2 NULL NULL
3 3 6DA20C06-0BAE-1609-FE7F-7054BF58367A Please enter 737950 into LINE within… 0 NULL 3 NULL NULL
4 4 88454216-2683-8567-C6EB-E471104D22B1 Please enter this code in the app to… 0 NULL 4 NULL NULL
5 5 0ACC60B1-1FEB-6038-CC1C-45B7ABDF9F03 Your Signal verification code: … 0 NULL 5 NULL NULL
6 6 F02C44BF-499E-7FEA-DC13-1C1993676D16 ThisIs, you have 2 new … 0 NULL 6 NULL NULL
7 7 B0A4B871-4EA8-ADCF-C03E-E6E6B5444EAC Telegram code: 95037… 0 NULL 7 NULL NULL
8 8 2A1B26B6-DE6E-E52C-44EA-FFE86D108C32 [TikTok] 8494 is your verification … 0 NULL 8 NULL NULL
9 9 33065BA2-CF1F-B2C6-D64D-B72CBB36D8DD Your WhatsApp code: 203-591… 0 NULL 9 NULL NULL
10 10 49790D66-CACD-CD04-1DF2-0D01094FB868 Your Skout verification code is 3079… 0 NULL 10 NULL NULL
11 11 61872292-B951-1A21-66B6-B2068262D0E0 Your Skout verification code is: … 0 NULL 11 NULL NULL
12 12 9868DE7C-0497-7FB6-A4C4-8C712AED2E53 ThisIs, you have 5 new … 0 NULL 6 NULL NULL
1 - 12 of 48
Go to: 1
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8

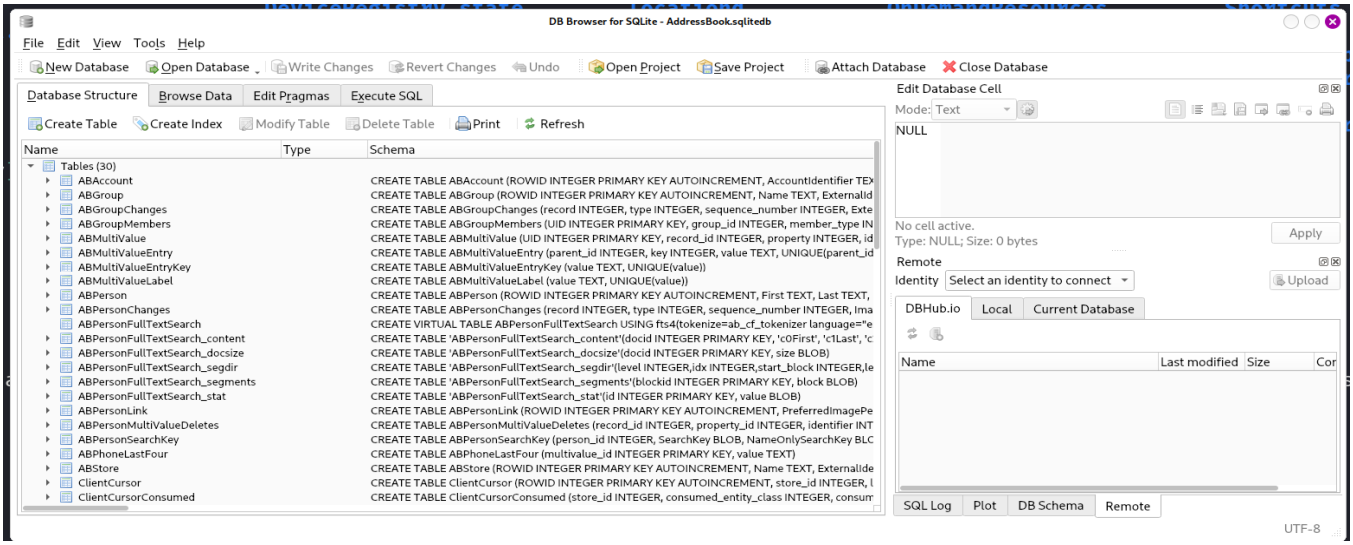
Addressbook.db

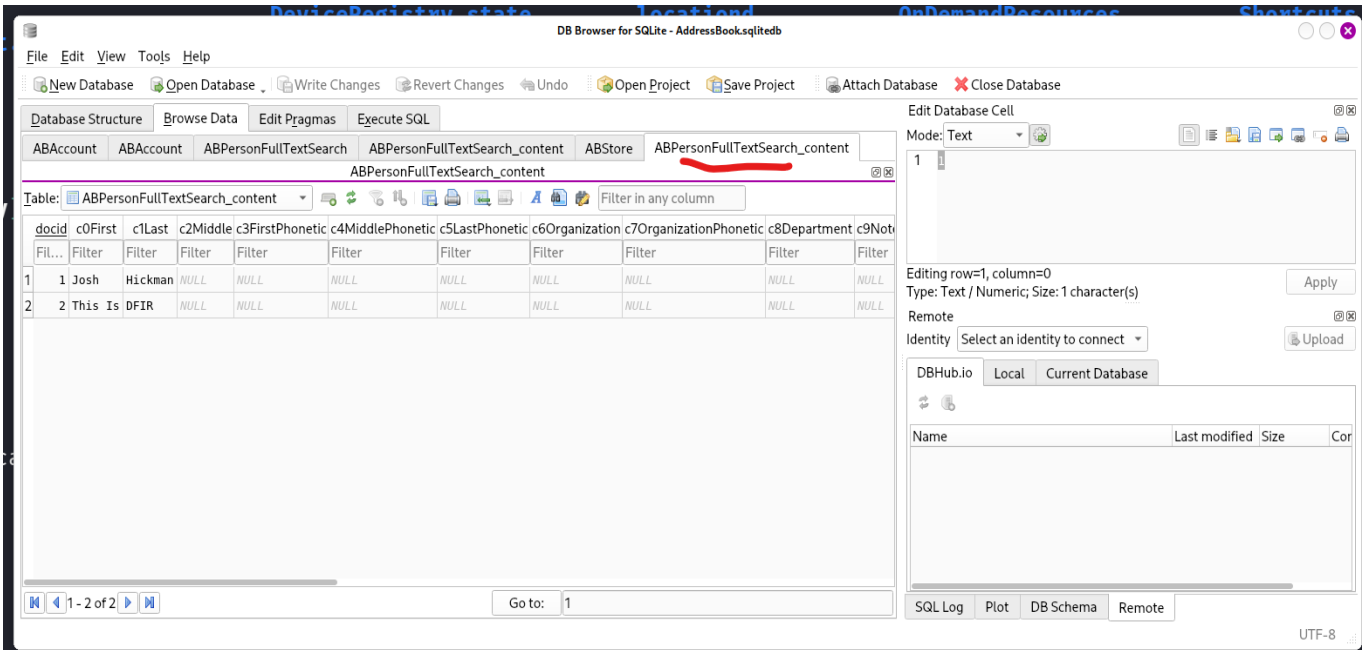
```
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ cd AddressBook

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/AddressBook]
└─$ ls
AddressBookImages.sqlitedb  AddressBookImages.sqlitedb-shm  AddressBookImages.sqlitedb-wal  AddressBook.sqlitedb  AddressBook.sqlitedb-shm  AddressBook.sqlitedb-wal

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/AddressBook]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/AddressBook]
└─$ sqlitebrowser AddressBook.sqlitedb
1.81-88: syntax error, unexpected identifier, expecting ) or ","
Sqlite parse error: CREATE VIRTUAL TABLE ABPersonFullTextSearch USING fts4(tokenize=ab_cf_tokenizer language="en-US" collator="en-US@collation=search", First, Last, Middle, FirstPhonetic, MiddlePhonetic, LastPhonetic, Org
anization, OrganizationPhonetic, Department, Note, Birthday, JobTitle, Nickname, Prefix, Suffix, DisplayName, Phone, Email, Address, SocialProfile, URL, RelatedNames, IM, Date, SupplementalCompositeNameTerms)
```





Task 4 Safari and calendar

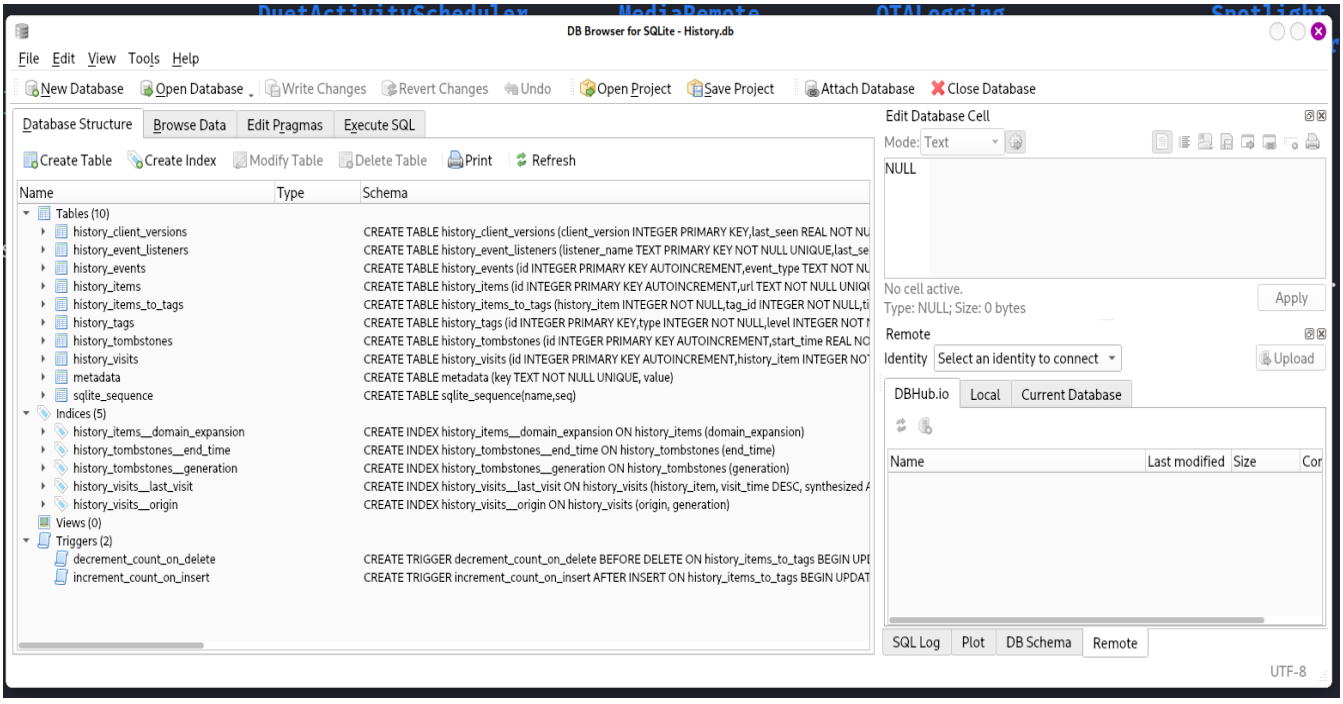
1. Examine Safari History.db and export at least five visits with URLs, titles, raw times and UTC conversions. Distinguish recorded visits from bookmarks, open tabs and synced tabs.

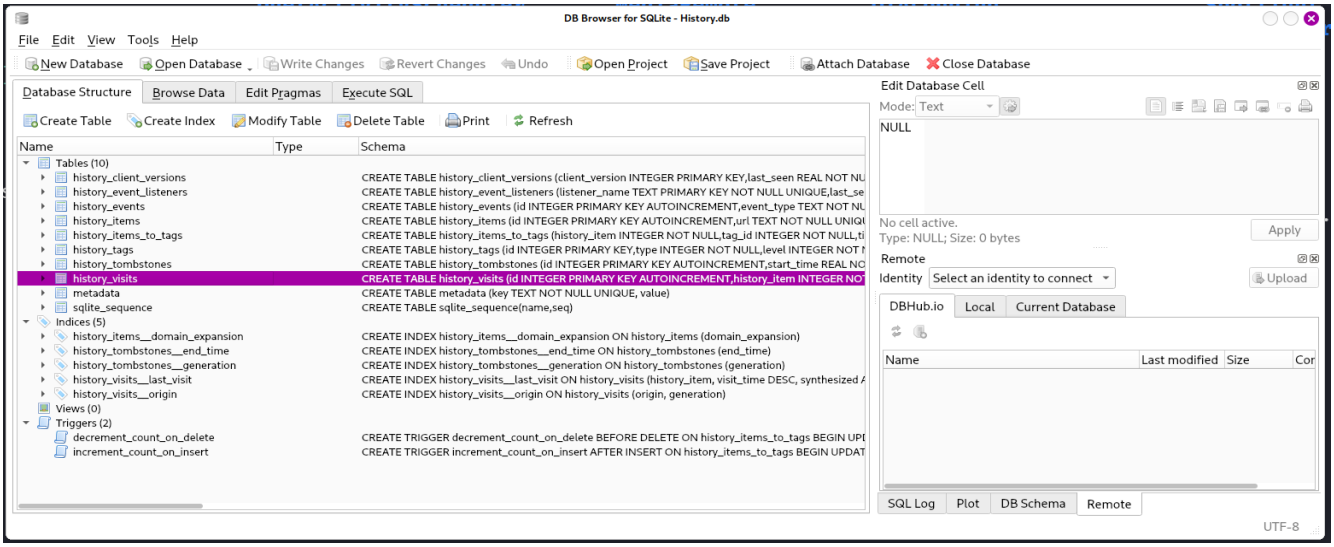
2. Inspect Calendar.sqlitedb and Extras.db where present. Record relevant tables and row counts, then report any supported event details. Document empty tables without inventing appointments.

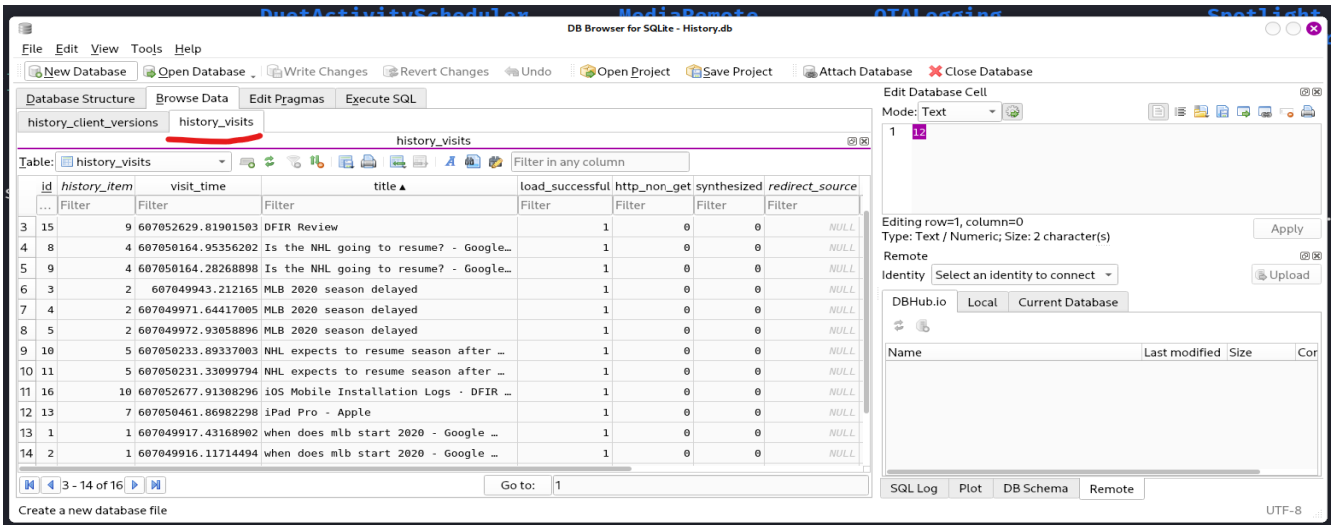
Safari History.db



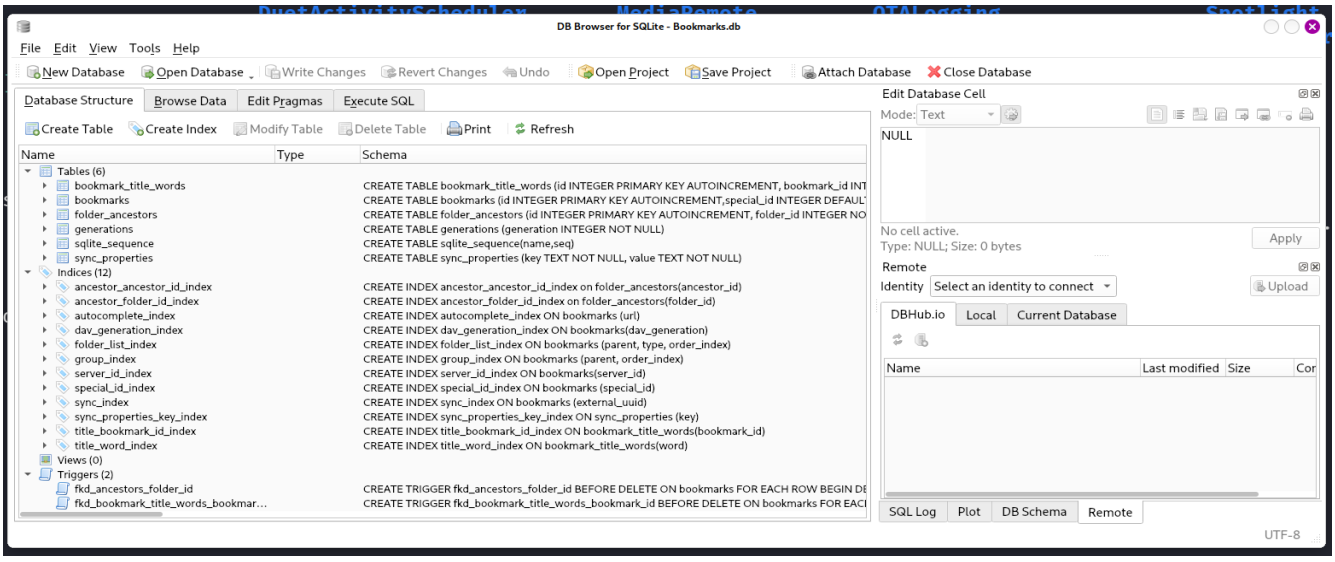




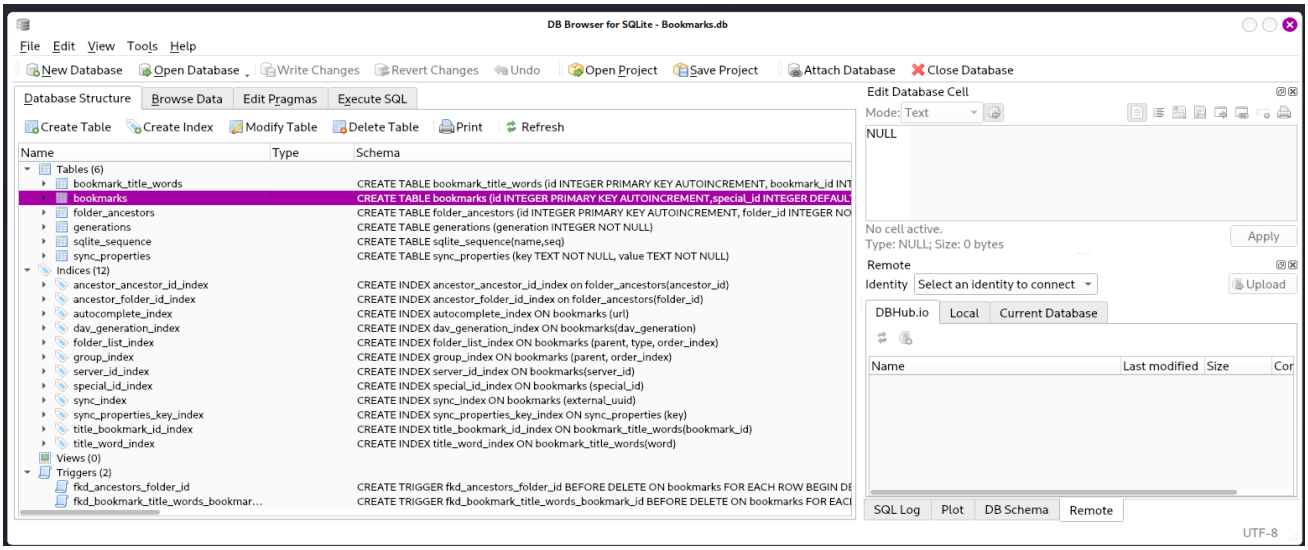

DB Browser for SQLite - History.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (10)
history_client_versions
history_event_listeners
history_events
history_items
history_items_to_tags
history_tags
history_tombstones
history_visits
metadata
sqlite_sequence
Indices (5)
history_items__domain_expansion
history_tombstones__end_time
history_tombstones__generation
history_visits__last_visit
history_visits__origin
Views (0)
Triggers (2)
decrement_count_on_delete
increment_count_on_insert
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8

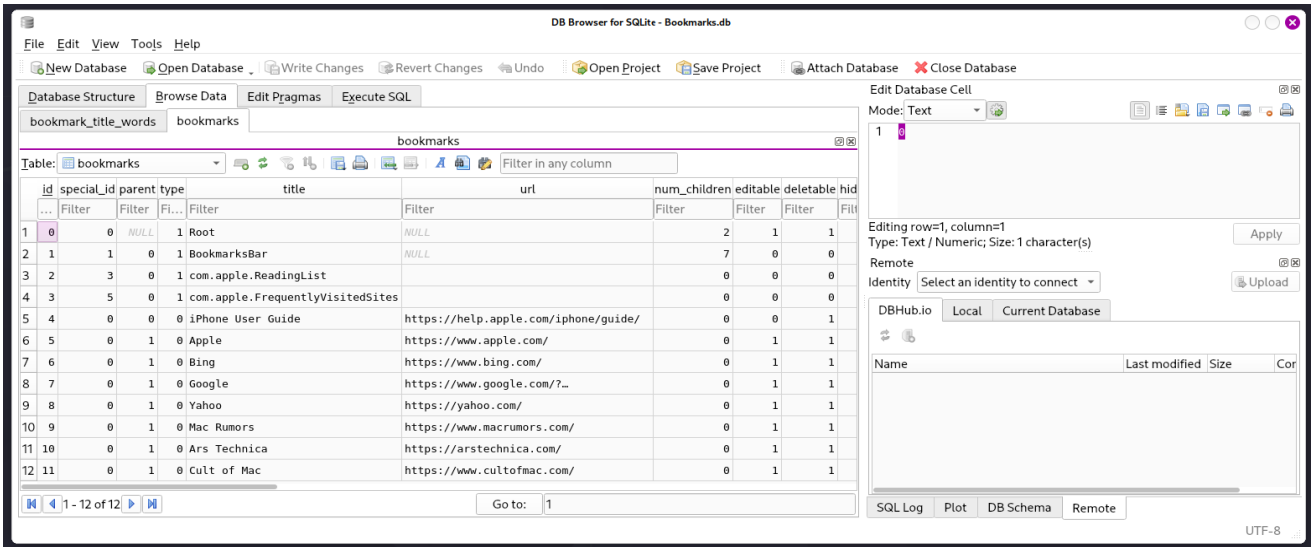

DB Browser for SQLite - History.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
history_client_versions
history_visits
history_visits
Table: history_visits
Filter in any column
id
history_item
visit_time
title
load_successful
http_non_get
synthesized
redirect_source
3 - 14 of 16
Go to: 1
Create a new database file
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 2 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8

Bookmark.db


DB Browser for SQLite - Bookmarks.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (6)
bookmark_title_words
bookmarks
folder_ancestors
generations
sqlite_sequence
sync_properties
Indices (12)
ancestor_ancestor_id_index
ancestor_folder_id_index
autocomplete_index
dav_generation_index
folder_list_index
group_index
server_id_index
special_id_index
sync_index
sync_properties_key_index
title_bookmark_id_index
title_word_index
Views (0)
Triggers (2)
fkd_ancestors_folder_id
fkd_bookmark_title_words_bookmar...
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Safari]
└─$ sqlitebrowser Bookmarks.db
```





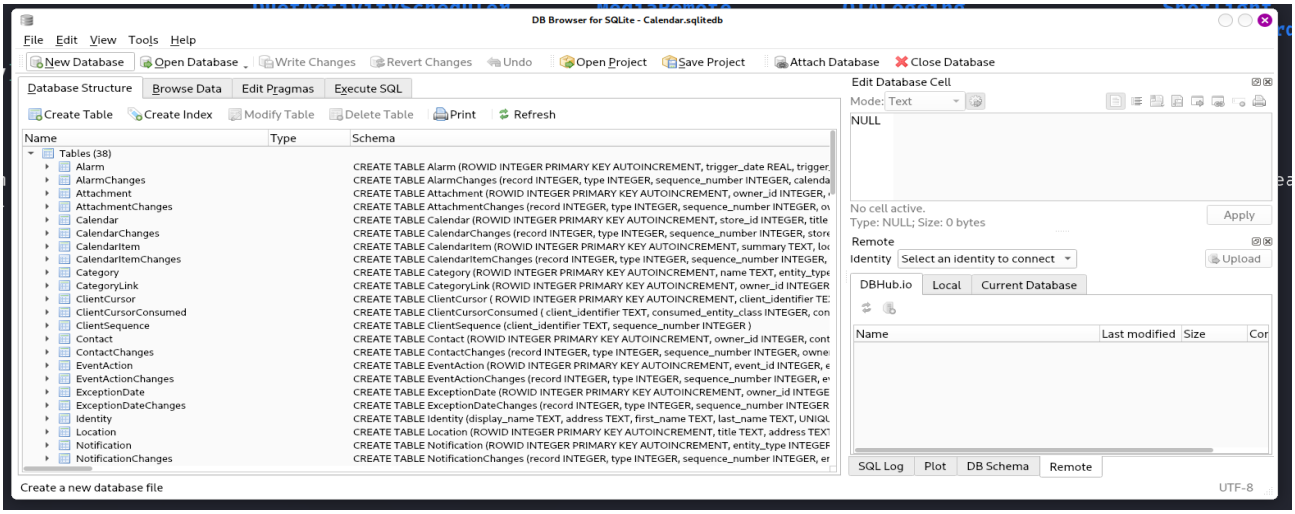
Calender and Extras

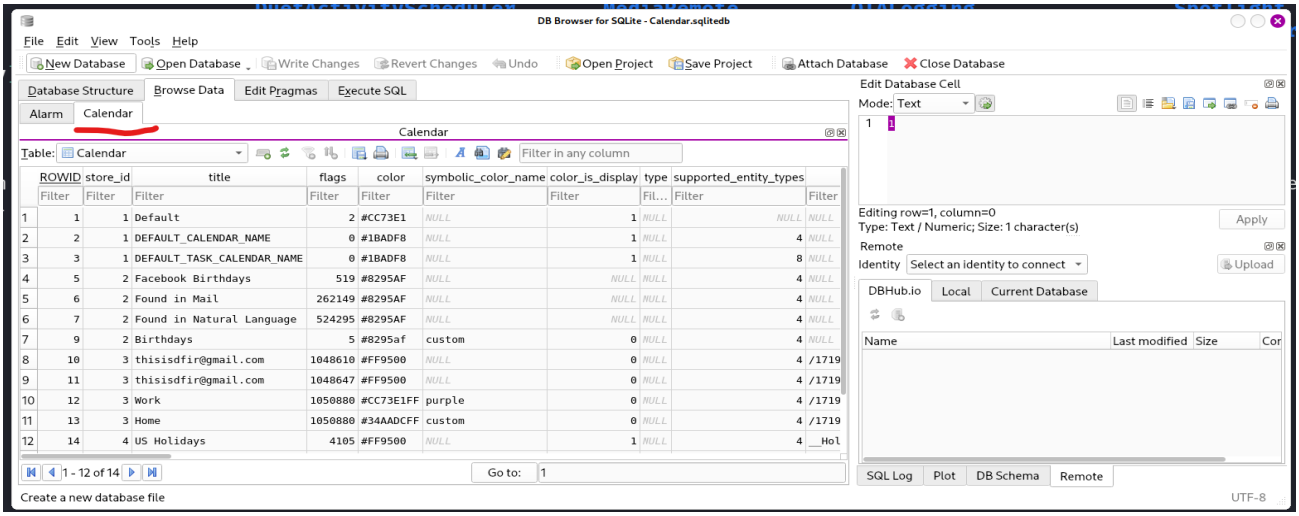
```
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ cd Calendar

┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Calendar]
└─$ ls
Calendar.sqlitedb      Calendar.sqlitedb-wal  Extras.db-shm  Notifications.Calendar.Protected                    Notifications.Calendar.TriggeredEvent.TimeToLeaveRefresh
Calendar.sqlitedb-shm  Extras.db              Extras.db-wal  Notifications.Calendar.TriggeredEvent.NotificationData  Notifications.Calendar.Unprotected

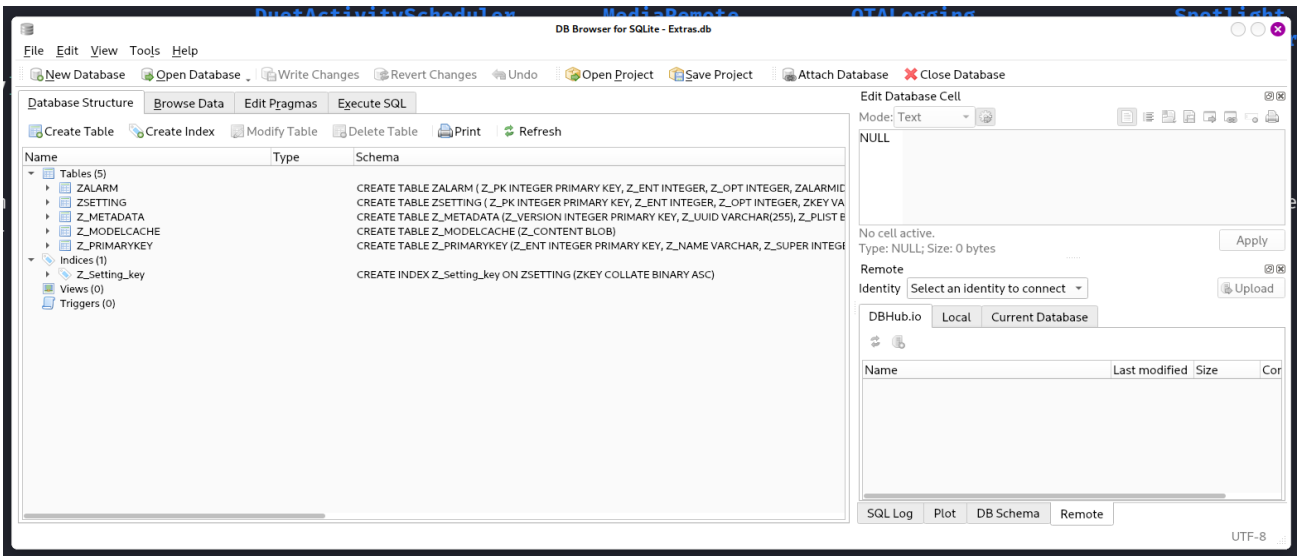
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Calendar]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Calendar]
└─$ sqlitebrowser Calendar.sqlitedb
```






```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Calendar]
└─$ sqlitebrowser Extras.db
```



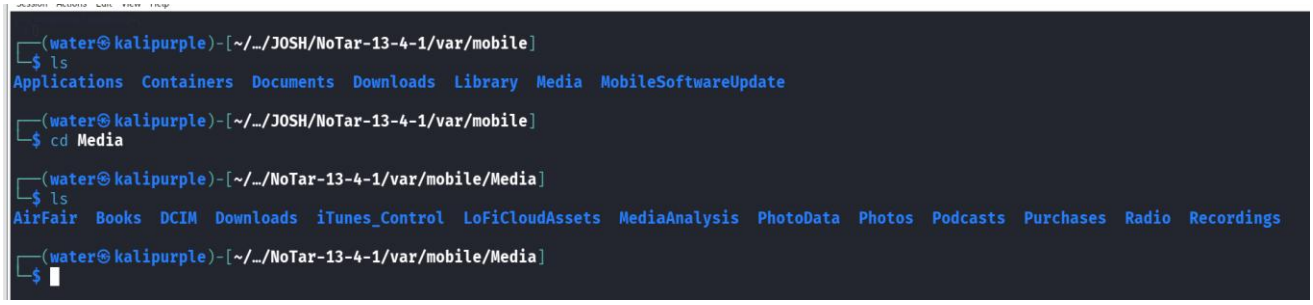
Task 5 Photographs and media

1. Examine Photos.sqlite, including ZGENERICASSET and related metadata tables as present. Select at least three media records and locate their corresponding files where available.

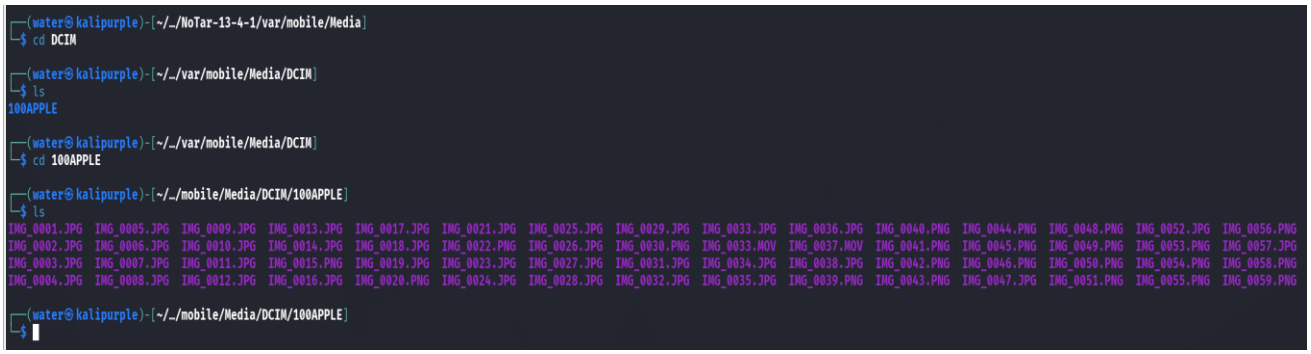
2. Record filenames, paths, hashes, capture-related and added times, available offsets and GPS metadata. Distinguish media captured by the camera from imported or saved media using supporting fields.

3. Validate one database result against file metadata. Explain discrepancies and any supported sharing or deletion indicators without assuming an indicator proves a complete transfer or deliberate deletion.

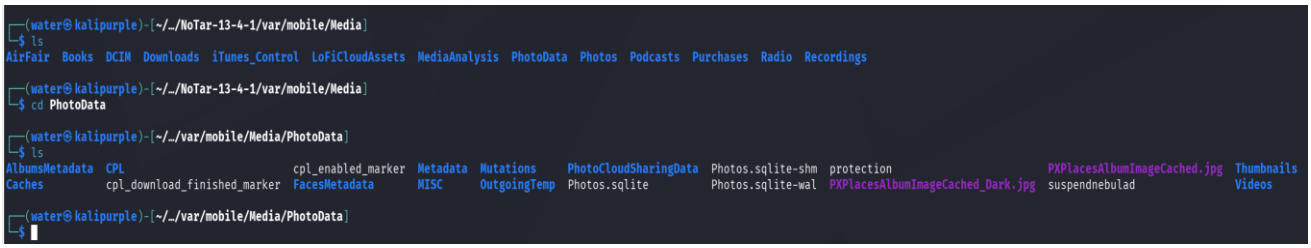
Media/Photo

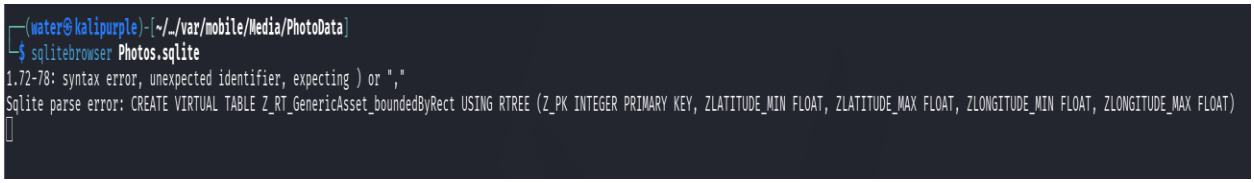


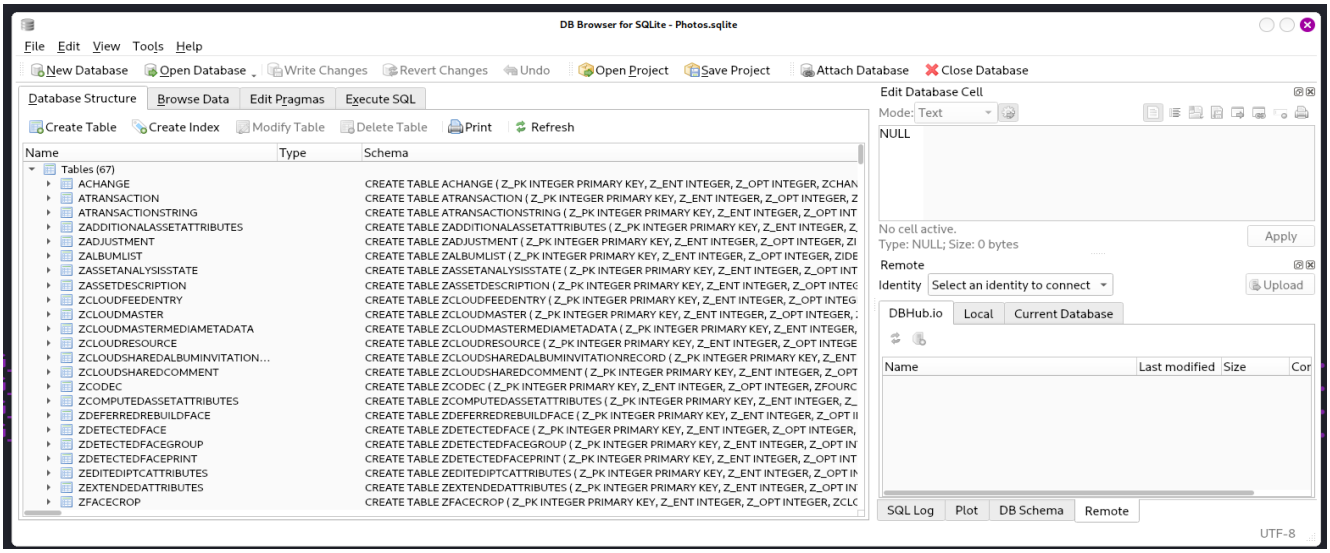
DCIM images

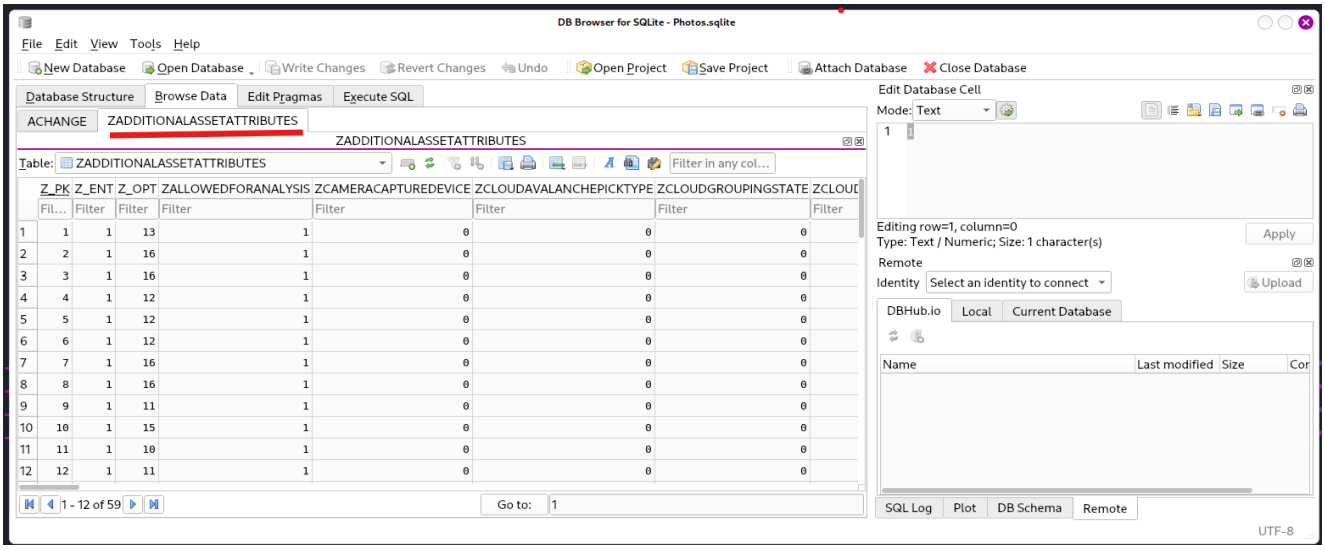


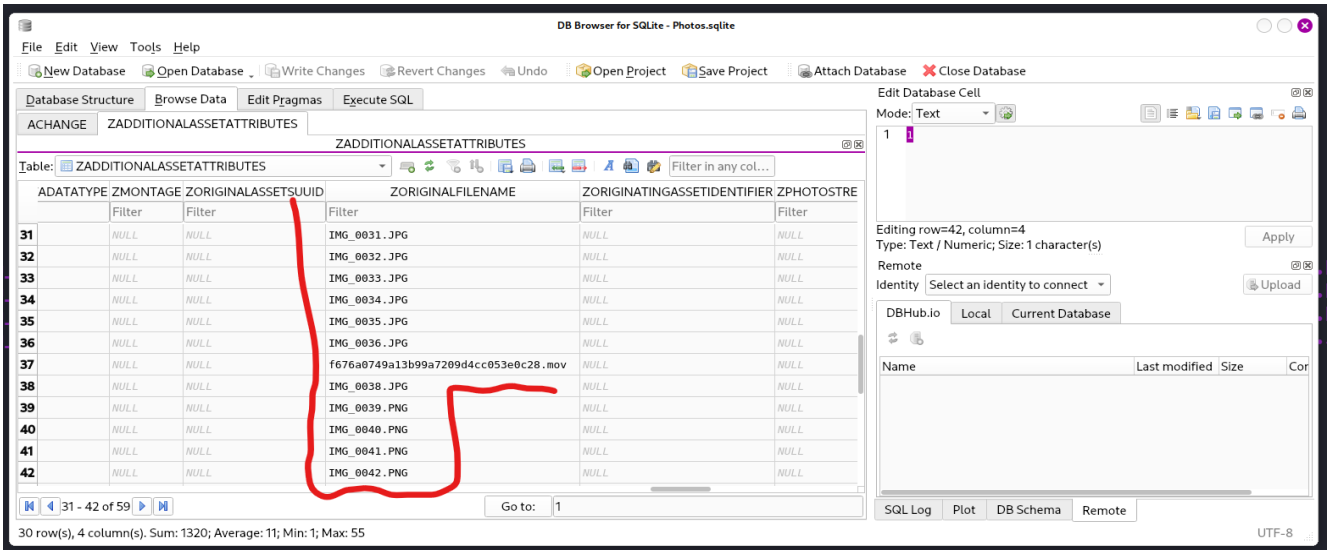
Photodata

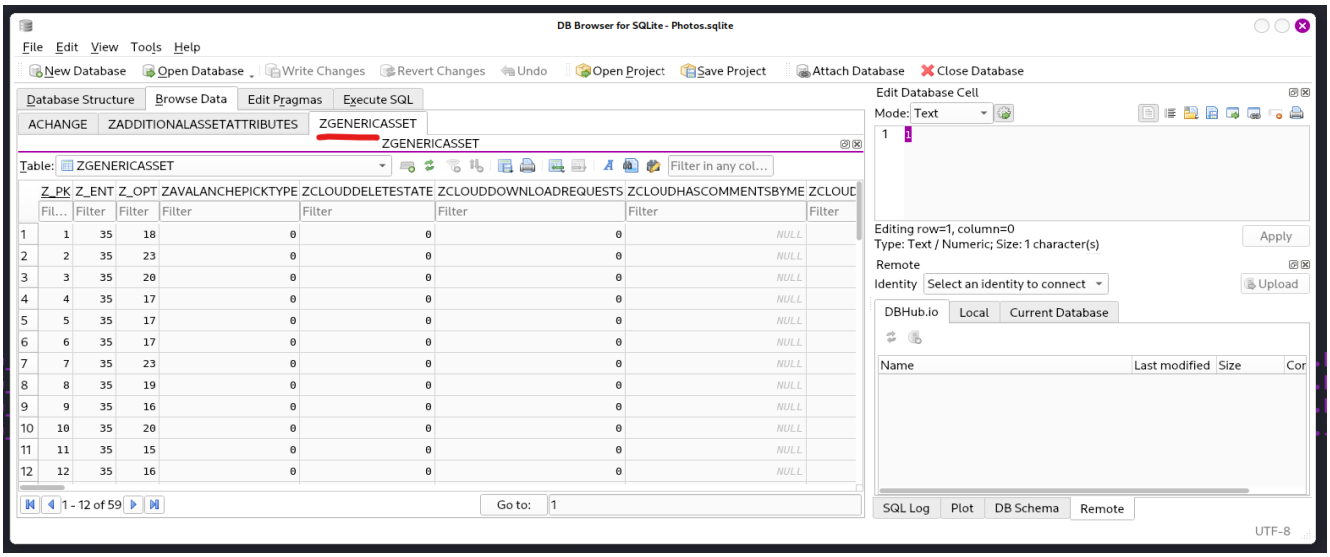


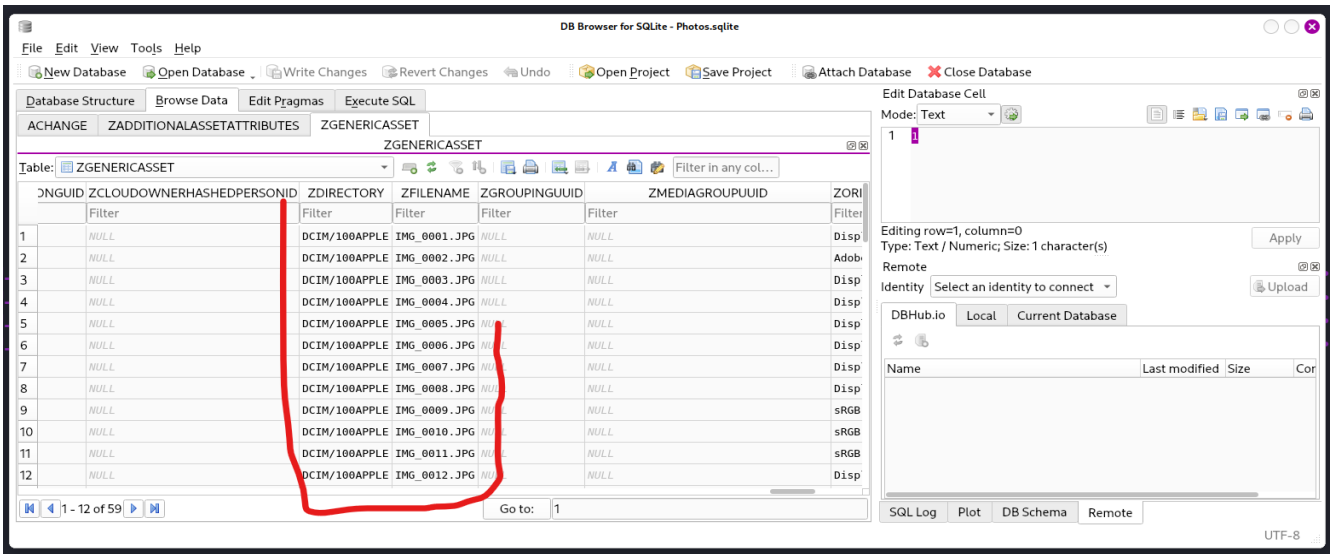



DB Browser for SQLite - Photos.sqlite
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
Create Table Create Index Modify Table Delete Table Print Refresh
Name Type Schema
Tables (67)
ACHANGE CREATE TABLE ACHANGE (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZCHAN
ATRANSACTION CREATE TABLE ATRANSACTION (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, Z
ATRANSACTIONSTRING CREATE TABLE ATRANSACTIONSTRING (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INT
ZADDITIONALASSETATTRIBUTES CREATE TABLE ZADDITIONALASSETATTRIBUTES (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z
ZADJUSTMENT CREATE TABLE ZADJUSTMENT (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZI
ZALBUMLIST CREATE TABLE ZALBUMLIST (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZIDE
ZASSETANALYSISSTATE CREATE TABLE ZASSETANALYSISSTATE (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INT
ZASSETDESCRIPTION CREATE TABLE ZASSETDESCRIPTION (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEG
ZCLOUDFEEDENTRY CREATE TABLE ZCLOUDFEEDENTRY (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEG
ZCLOUDMASTER CREATE TABLE ZCLOUDMASTER (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER,
ZCLOUDMASTERMEDIAMETADATA CREATE TABLE ZCLOUDMASTERMEDIAMETADATA (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER,
ZCLOUDRESOURCE CREATE TABLE ZCLOUDRESOURCE (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGE
ZCLOUDSHAREDALBUMINVITATION... CREATE TABLE ZCLOUDSHAREDALBUMINVITATIONRECORD (Z_PK INTEGER PRIMARY KEY, Z_ENT
ZCLOUDSHAREDCOMMENT CREATE TABLE ZCLOUDSHAREDCOMMENT (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT
ZCODEC CREATE TABLE ZCODEC (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZFOURC
ZCOMPUTEDASSETATTRIBUTES CREATE TABLE ZCOMPUTEDASSETATTRIBUTES (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_
ZDEFERREDREBUILDFACE CREATE TABLE ZDEFERREDREBUILDFACE (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT I
ZDETECTEDFACE CREATE TABLE ZDETECTEDFACE (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER,
ZDETECTEDFACEGROUP CREATE TABLE ZDETECTEDFACEGROUP (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT IN
ZDETECTEDFACEPRINT CREATE TABLE ZDETECTEDFACEPRINT (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INT
ZEDITEDIPTCATTRIBUTES CREATE TABLE ZEDITEDIPTCATTRIBUTES (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT I
ZEXTENDEDATTRIBUTES CREATE TABLE ZEXTENDEDATTRIBUTES (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT IN
ZFACECROP CREATE TABLE ZFACECROP (Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZCLC
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8


DB Browser for SQLite - Photos.sqlite
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
ACHANGE ZADDITIONALASSETATTRIBUTES
ZADDITIONALASSETATTRIBUTES
Table: ZADDITIONALASSETATTRIBUTES
Filter in any col...
Z_PK Z_ENT Z_OPT ZALLOWEDFORANALYSIS ZCAMERACAPTUREDEVICE ZCLOUDAVALANCHEPICKTYPE ZCLOUDGROUPINGSTATE ZCLOUI
1 1 1 13 1 0 0 0
2 2 1 16 1 0 0 0
3 3 1 16 1 0 0 0
4 4 1 12 1 0 0 0
5 5 1 12 1 0 0 0
6 6 1 12 1 0 0 0
7 7 1 16 1 0 0 0
8 8 1 16 1 0 0 0
9 9 1 11 1 0 0 0
10 10 1 15 1 0 0 0
11 11 1 10 1 0 0 0
12 12 1 11 1 0 0 0
1 - 12 of 59
Go to: 1
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8


DB Browser for SQLite - Photos.sqlite
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
ACHANGE ZADDITIONALASSETATTRIBUTES
ZADDITIONALASSETATTRIBUTES
Table: ZADDITIONALASSETATTRIBUTES
Filter in any col...
ADATATYPE ZMONTAGE ZORIGINALASSETSUUID ZORIGINALFILENAME ZORIGINATINGASSETIDENTIFIER ZPHOTOSTRE
31 NULL NULL IMG_0031.JPG NULL NULL
32 NULL NULL IMG_0032.JPG NULL NULL
33 NULL NULL IMG_0033.JPG NULL NULL
34 NULL NULL IMG_0034.JPG NULL NULL
35 NULL NULL IMG_0035.JPG NULL NULL
36 NULL NULL IMG_0036.JPG NULL NULL
37 NULL NULL f676a0749a13b99a7209d4cc053e0c28.mov NULL NULL
38 NULL NULL IMG_0038.JPG NULL NULL
39 NULL NULL IMG_0039.PNG NULL NULL
40 NULL NULL IMG_0040.PNG NULL NULL
41 NULL NULL IMG_0041.PNG NULL NULL
42 NULL NULL IMG_0042.PNG NULL NULL
31 - 42 of 59
Go to: 1
30 row(s), 4 column(s). Sum: 1320; Average: 11; Min: 1; Max: 55
Edit Database Cell
Mode: Text
Editing row=42, column=4
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity Select an identity to connect
Upload
DBHub.io Local Current Database
Name Last modified Size Cor
SQL Log Plot DB Schema Remote
UTF-8





Task 6 KnowledgeC activity

1. Locate private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db. Inspect ZOBJECT and related metadata, then extract at least five relevant records, such as application focus or device lock activity.

2. Report the stream name, record ID, associated application or value, raw start and end times, UTC conversions and duration where meaningful. Correlate at least one interval with messages, browsing or media activity.

```
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Media]
└─$ cd ..

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/mobile]
└─$ ls
Applications  Containers  Documents  Downloads  Library  Media  MobileSoftwareUpdate

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/mobile]
└─$ cd Library
```

```
┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/mobile]
└─$ cd Library

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ ls
Accessibility            'Carrier Bundle.bundle'                  Cookies                  ExternalAccessory     MedicalID                PersonalizationPortrait  Suggestions
Accounts                 'Carrier Bundles'                        CoreAccessories          FairPlay              MessagesMetaData         PowerUI                  SyncedPreferences
AddressBook               CarrierDefault.bundle                   CoreAS                   FrontBoard            MobileContainerManager   Preferences              TCC
AggregateDictionary       ClassKit                                CoreBehavior             GameKit              'Mobile Documents'        Recents                  terminus
AirPlayRoutePrediction    com.apple.assistant.Contacts            CoreDuet                 Health                MobileInstallation       RelevanceEngine          Trial
AppConduit                com.apple.companionappd                 CoreFollowUp             homed                 MusicLibrary             Reminders                UserConfigurationProfiles
ApplePushService          com.apple.icloud.searchpartyd           coreidvd                 IdentityServices      NanoMusicSync            ReplayKit                UserFonts
'Application Support'     com.apple.internal.ck                   CrashReporter            Inboxes               NanoPhotos               ResponseKit              UserNotifications
Assistant                 com.apple.iTunesCloud                   Cydia                    Intents               NanoPreferencesSync      Safari                   UserNotificationsServer
Avatar                    com.apple.itunesstored                  DataAccess               Keyboard              NanoTimeKit              SafariSafeBrowsing       Voicemail
BackBoard                 com.apple.nsurlsessiond                 DES                      KeyboardServices      News                    'Saved Application State' VoiceServices
BulletinBoard             com.apple.nsurlsessiond-launchd         DeviceRegistry           LASD                  NFStorage                SecureElementService     VoiceTrigger
Caches                    com.apple.sensorkitd                    DeviceRegistry.dontBackUp Library              Notes                    Sharing                  WatchConnectivity
Calendar                  com.apple.siri.inference                DeviceRegistry.state     locationd             OnDemandResources        Shortcuts                WebClips
CallDirectory             com.apple.VideoSubscriberAccount.DeveloperService DifferentialPrivacy Logs              Operator1Bundle.bundle   SMS                      WebKit
CallHistoryDB             com.apple.WatchListKit                  dmd                      mad                  'Operator Bundle.bundle'  SoftwareUpdate           WelcomeMat
CallHistoryTransactions   Contacts                                DoNotDisturb             Mail                  OSAnalytics              SplashBoard
CallServices              ContactsMetadata                        DuetActivityScheduler    MediaRemote           OTALogging               Spotlight
CarrierBundle.bundle      ControlCenter                           DuetExpertCenter         MediaStream           Passes                   SpringBoard

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ cd CoreDuet

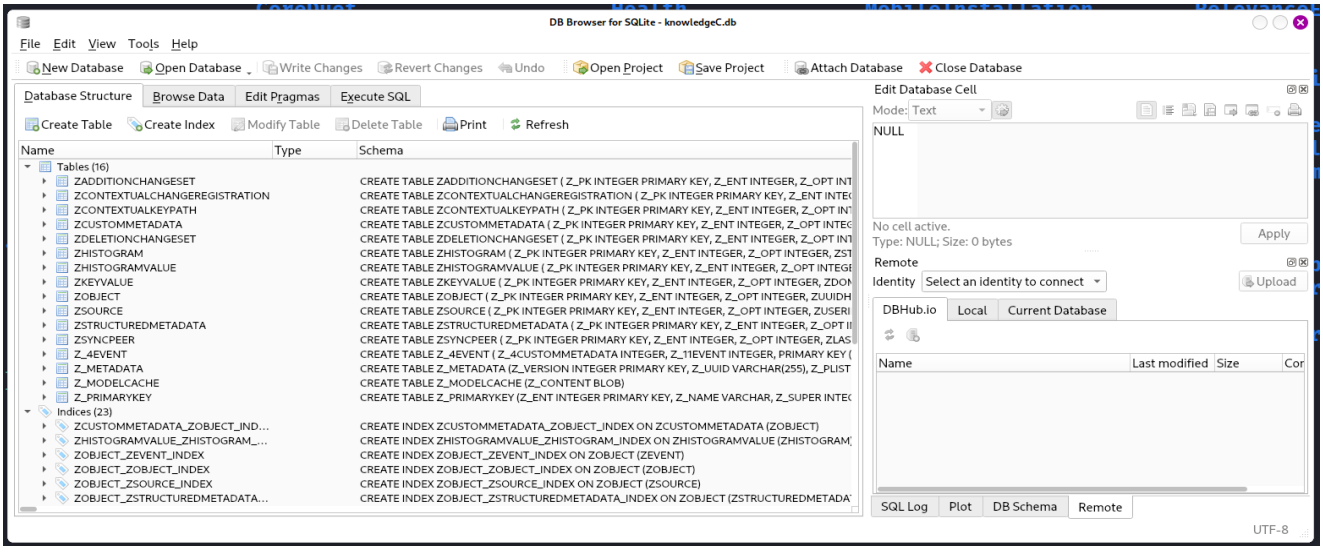
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/CoreDuet]
└─$ ls
Knowledge  People

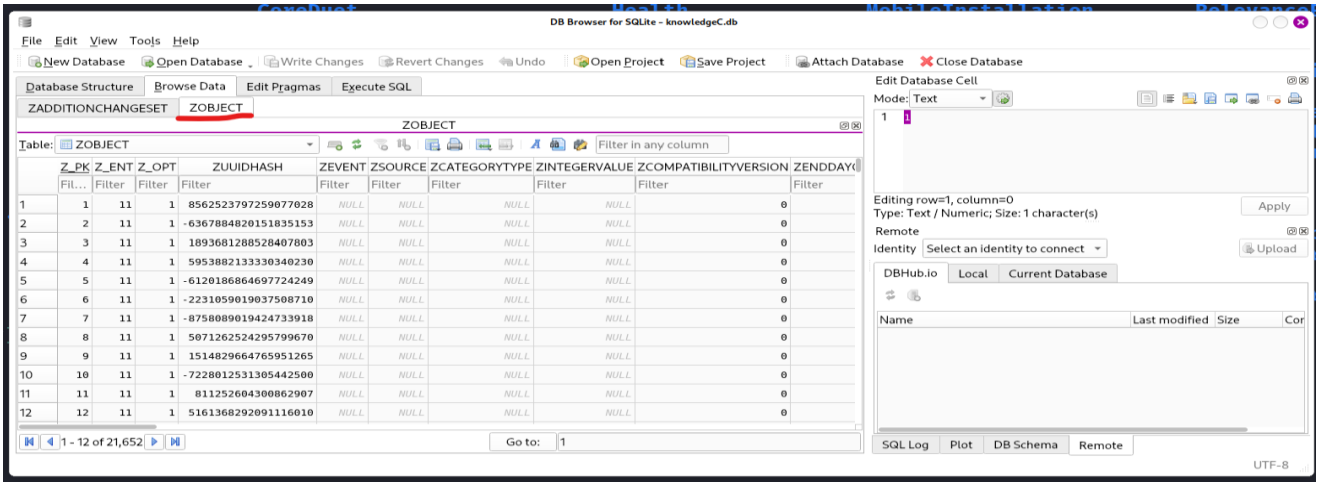
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/CoreDuet]
└─$ cd Knowledge

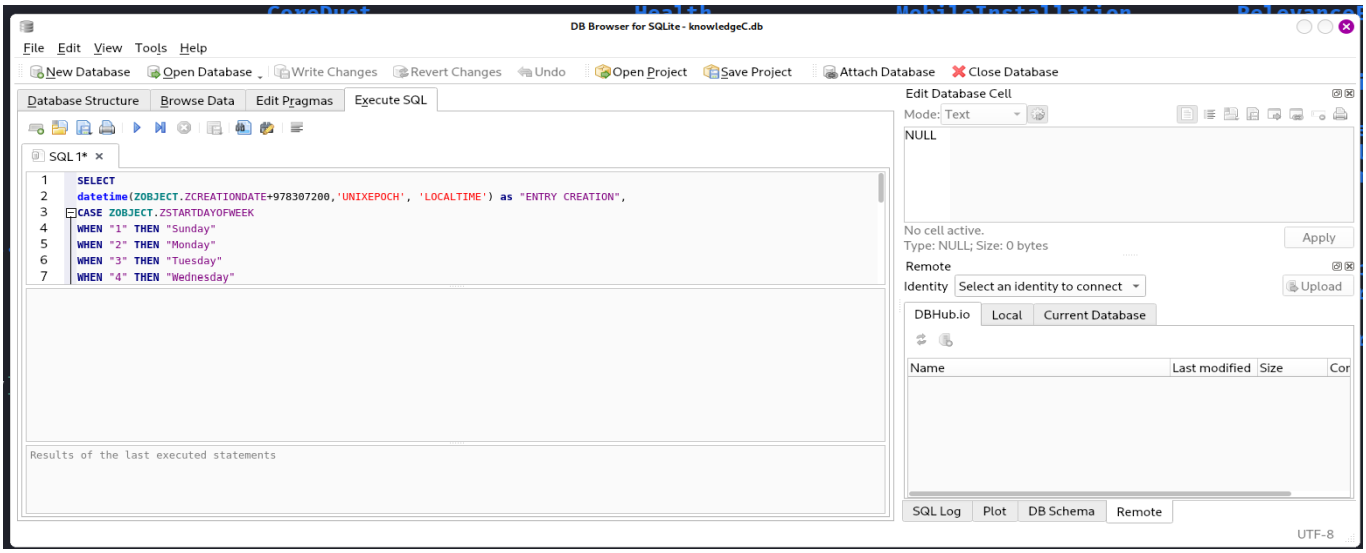
┌──(water㉿kalipurple)-[~/…/mobile/Library/CoreDuet/Knowledge]
└─$ ls
knowledgeC.db  knowledgeC.db-shm  knowledgeC.db-wal

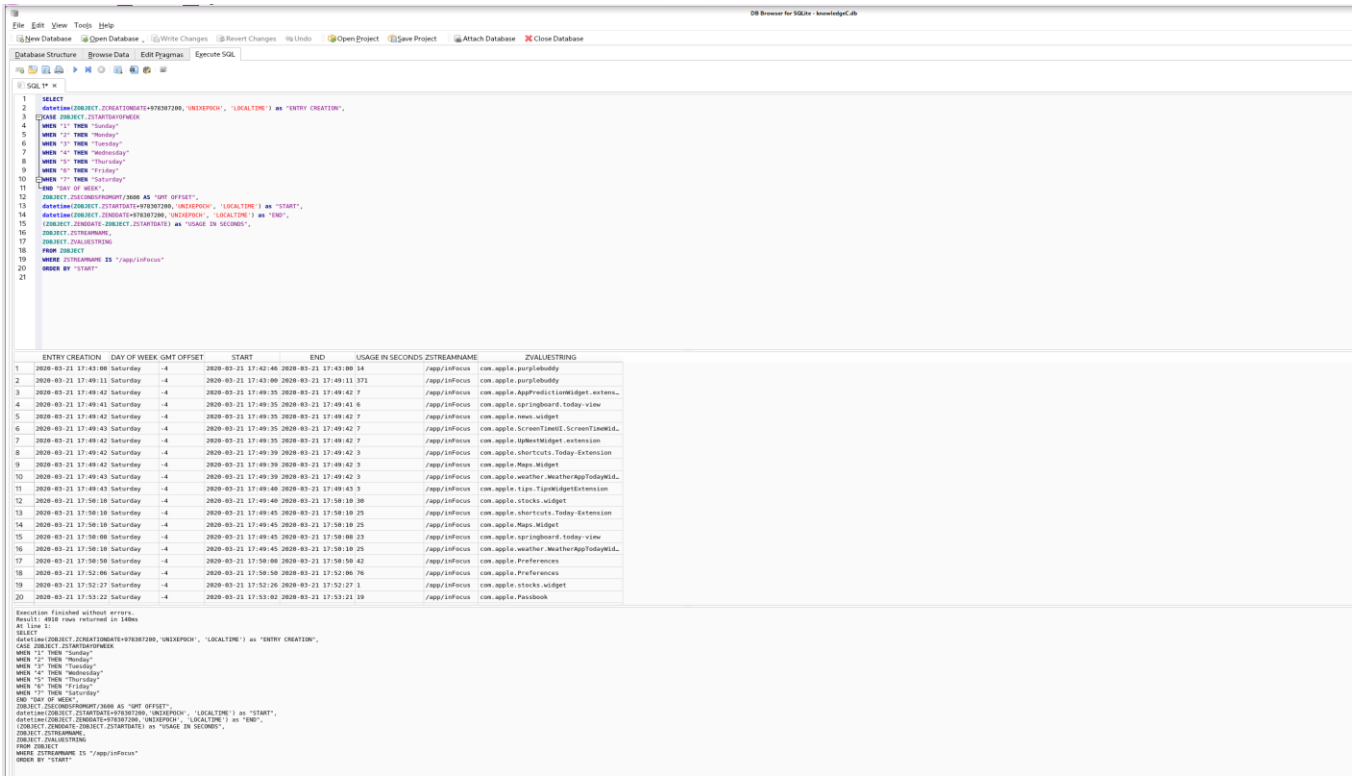
┌──(water㉿kalipurple)-[~/…/mobile/Library/CoreDuet/Knowledge]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/mobile/Library/CoreDuet/Knowledge]
└─$ sqlitebrowser knowledgeC.db
```









Task 7 Health and location evidence

1. Inspect healthdb.sqlite and healthdb_secure.sqlite where present. Identify a recorded data source or device and examine at least two activity samples. Report times, values and units without treating a synced wearable record as proof the phone holder performed the activity.
2. Inspect available routined Local.sqlite and locationd cache_encryptedB.db records. Examine at least three location-related entries where available, recording source tables, coordinates, times and their meanings.

3. Distinguish learned visits from Wi-Fi or cell reference locations. Use supported visit start and end fields where available; a record creation time is not automatically a visit time. Compare one location result with media or activity evidence.

Health

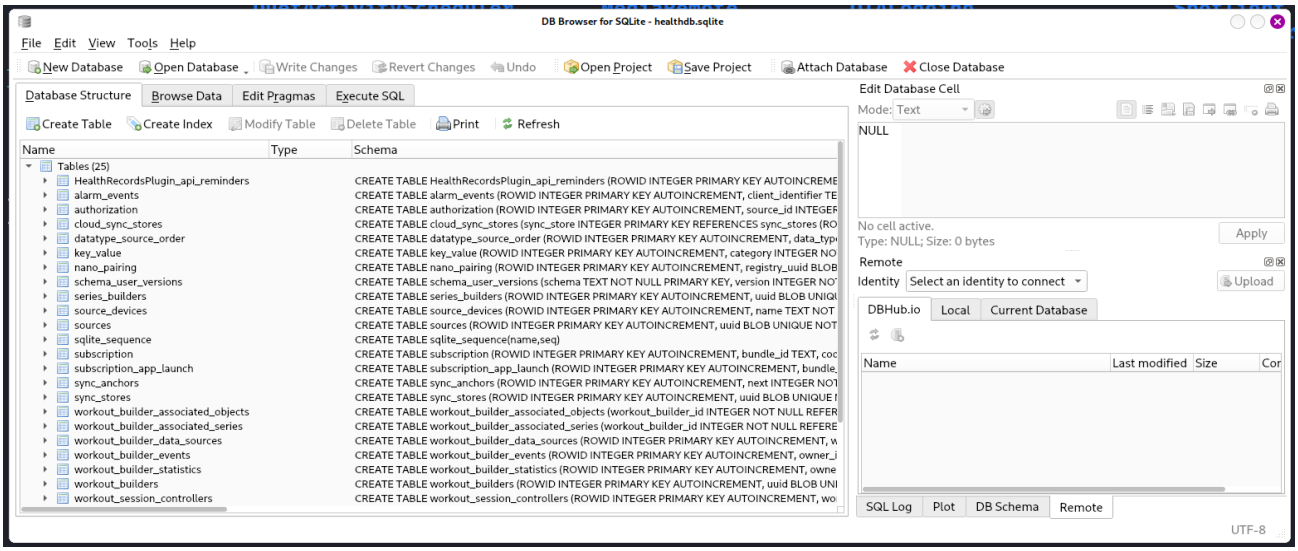
```
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ ls
Accessibility            'Carrier Bundle.bundle'                 Cookies                  ExternalAccessory   MedicalID                  PersonalizationPortrait   Suggestions
Accounts                 'Carrier Bundles'                       CoreAccessories          FairPlay            MessagesMetaData           PowerUI                   SyncedPreferences
AddressBook              CarrierDefault.bundle                   CoreAS                   FrontBoard          MobileContainerManager     Preferences               TCC
AggregateDictionary      ClassKit                                CoreBehavior             GameKit            'Mobile Documents'          Recents                   terminus
AirPlayRoutePrediction   com.apple.assistant.Contacts            CoreDuet                 Health              MobileInstallation         RelevanceEngine           Trial
AppConduit               com.apple.companionappd                 CoreFollowUp             homed               MusicLibrary               Reminders                 UserConfigurationProfiles
ApplePushService         com.apple.icloud.searchpartyd           coreidvd                 IdentityServices    NanoMusicSync              ReplayKit                 UserFonts
'Application Support'    com.apple.internal.ck                   CrashReporter            Inboxes             NanoPhotos                 ResponseKit               UserNotifications
Assistant                com.apple.iTunesCloud                   Cydia                    Intents             NanoPreferencesSync        Safari                    UserNotificationsServer
Avatar                   com.apple.itunesstored                  DataAccess               Keyboard            NanoTimeKit                SafariSafeBrowsing        Voicemail
BackBoard                com.apple.nsurlsessiond                 DES                      KeyboardServices    News                      'Saved Application State'  VoiceServices
BulletinBoard            com.apple.nsurlsessiond-launchd         DeviceRegistry           LASD                NFStorage                  SecureElementService      VoiceTrigger
Caches                   com.apple.sensorkitd                    DeviceRegistry.dontBackUp Library            Notes                      Sharing                   WatchConnectivity
Calendar                 com.apple.siri.inference                DeviceRegistry.state     locationd           OnDemandResources          Shortcuts                 WebClips
CallDirectory            com.apple.VideoSubscriberAccount.DeveloperService DifferentialPrivacy Logs            Operator1Bundle.bundle     SMS                       WebKit
CallHistoryDB            com.apple.WatchListKit                  dmd                      mad                'Operator Bundle.bundle'    SoftwareUpdate            WelcomeMat
CallHistoryTransactions  Contacts                                DoNotDisturb             Mail                OSAnalytics                SplashBoard
CallServices             ContactsMetadata                        DuetActivityScheduler    MediaRemote         OTALogging                 Spotlight
Carrier1Bundle.bundle    ControlCenter                           DuetExpertCenter         MediaStream         Passes                     SpringBoard

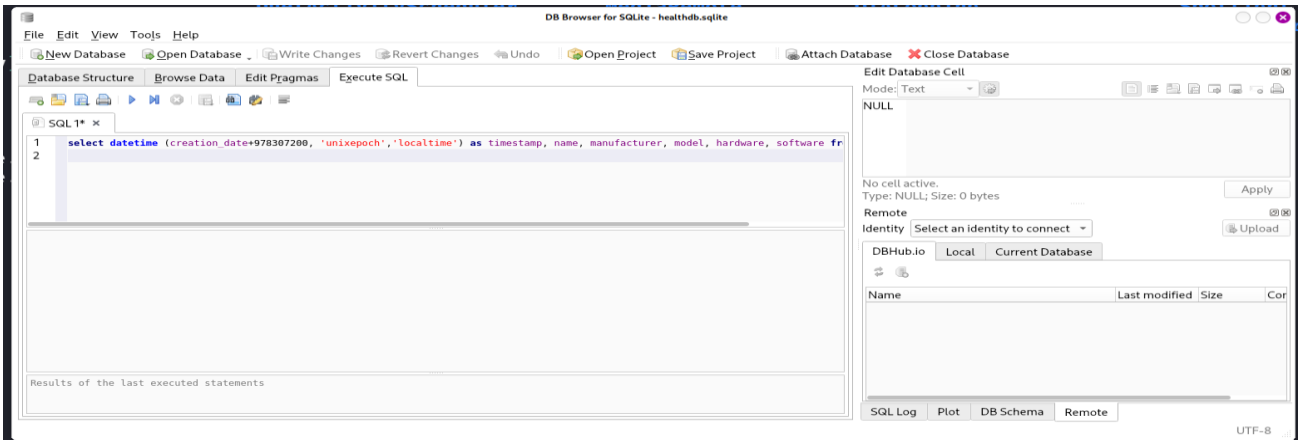
┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
└─$ cd Health

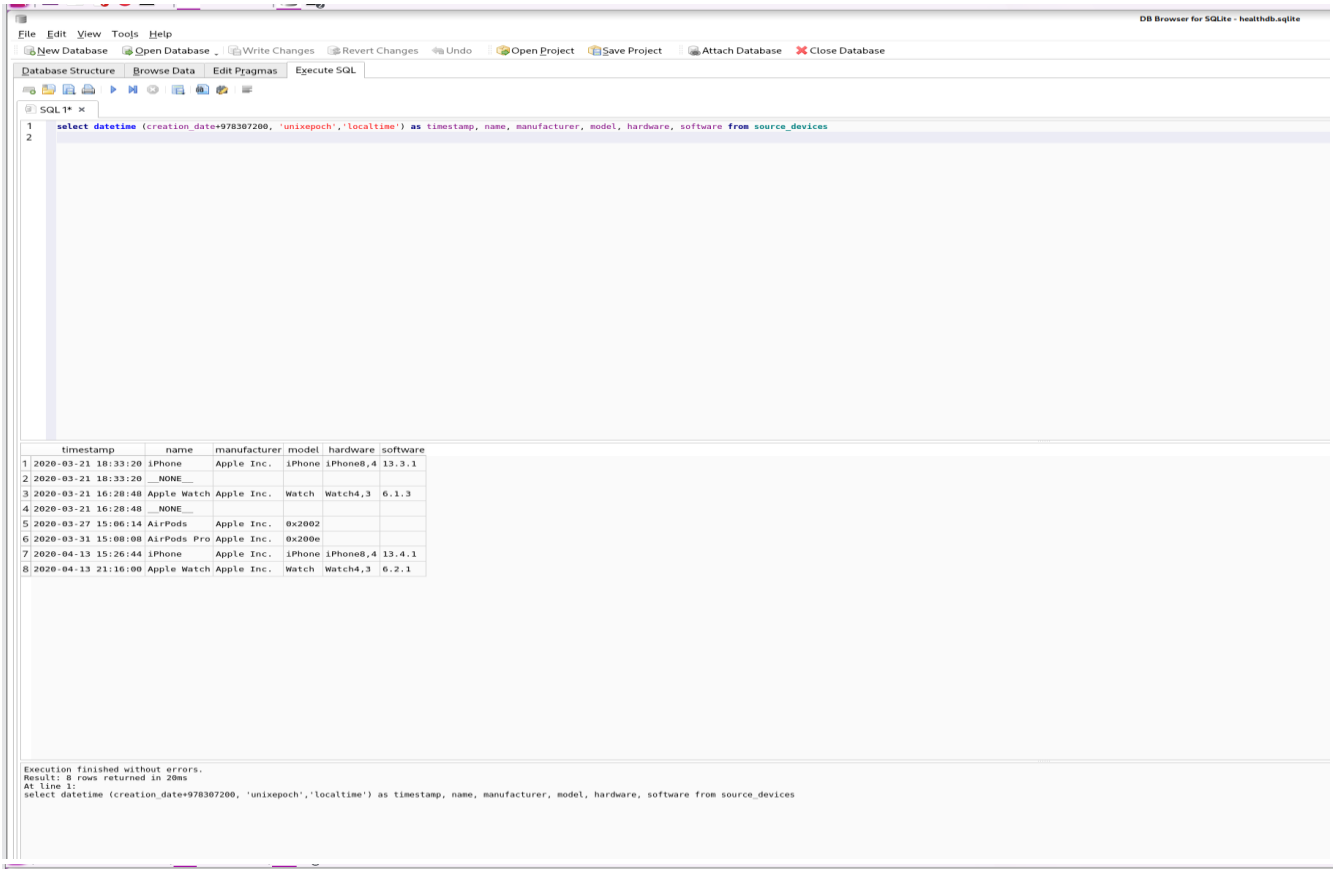
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Health]
└─$ ls
Client            healthdb_secure.hfd      healthdb_secure.sqlite      healthdb_secure.sqlite-wal  healthdb.sqlite-shm  Journals
CloudSyncJournals  healthdb_secure.hfd-wal  healthdb_secure.sqlite-shm  healthdb.sqlite             healthdb.sqlite-wal  NanoSync

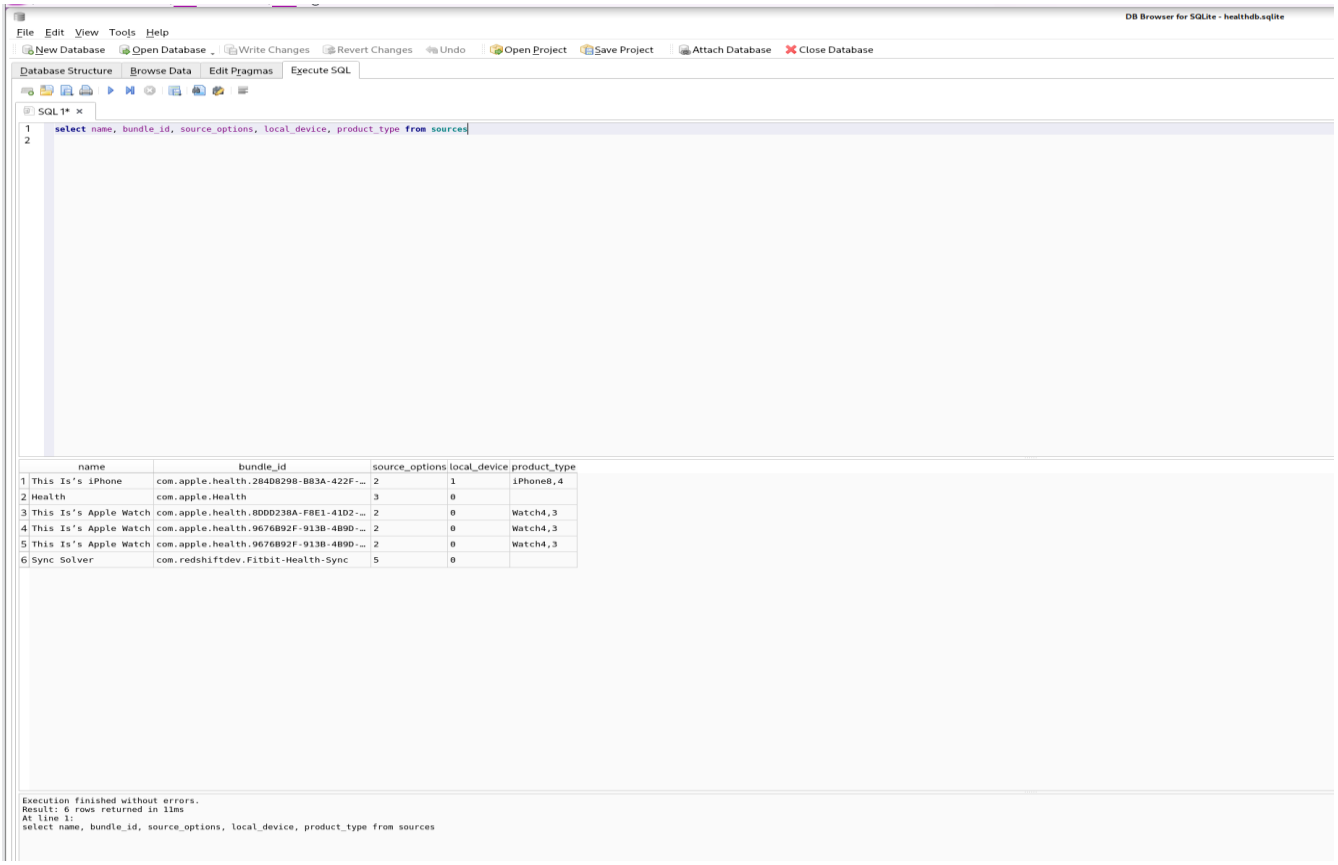
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Health]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Health]
└─$ sqlitebrowser healthdb.sqlite
```

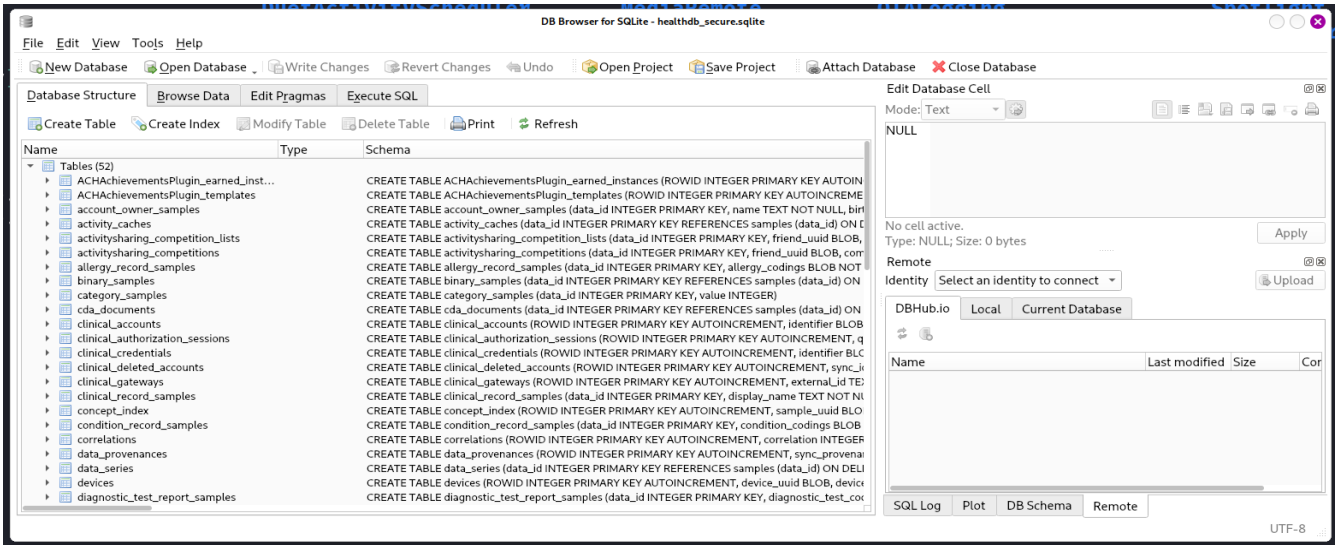


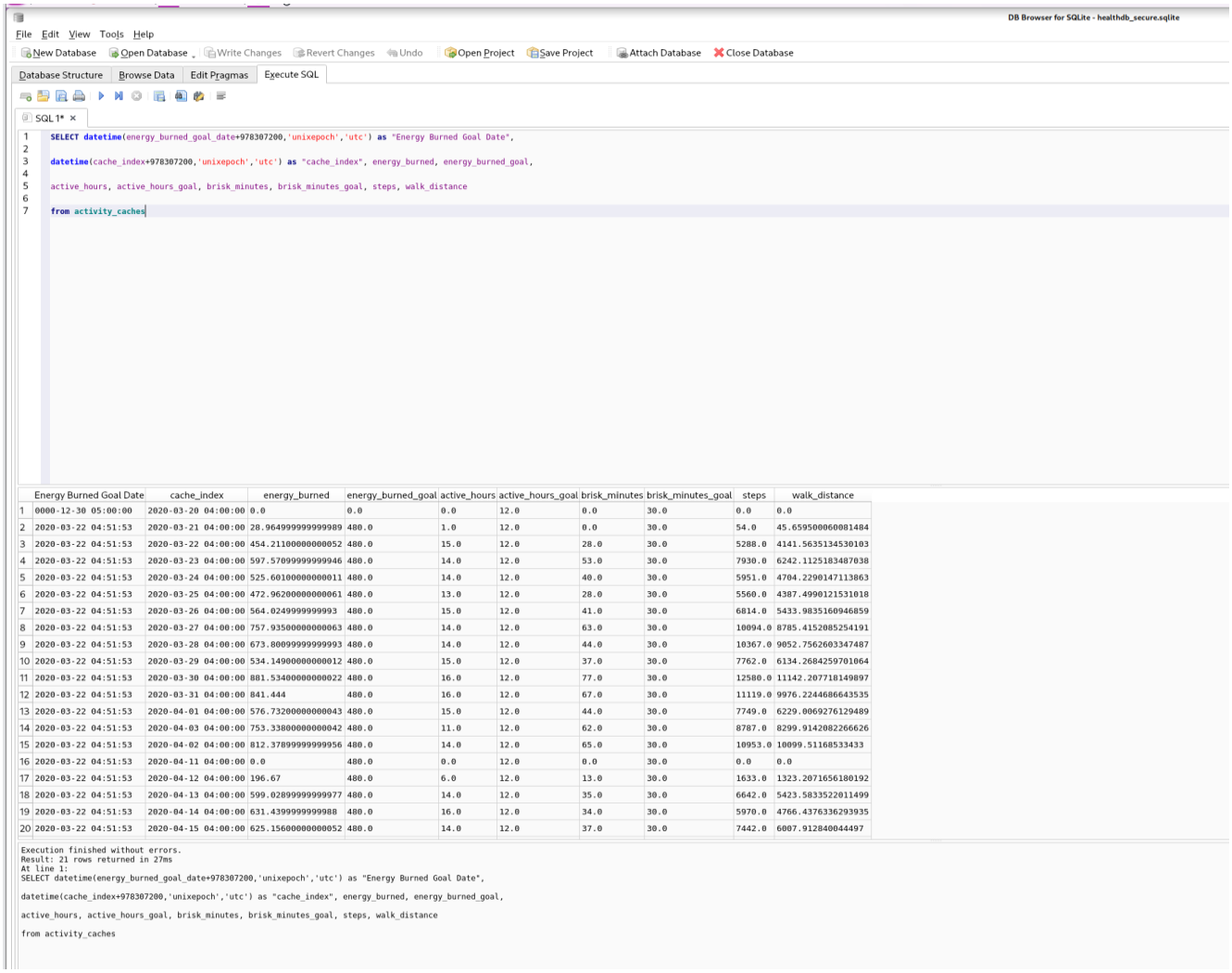



DB Browser for SQLite - healthdb.sqlite
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
SQL 1*
1 select datetime (creation_date+978307200, 'unixepoch','localtime') as timestamp, name, manufacturer, model, hardware, software from source_devices
2
timestamp name manufacturer model hardware software
1 2020-03-21 18:33:20 iPhone Apple Inc. iPhone iPhone8,4 13.3.1
2 2020-03-21 18:33:20 __NONE__
3 2020-03-21 16:28:48 Apple Watch Apple Inc. Watch Watch4,3 6.1.3
4 2020-03-21 16:28:48 __NONE__
5 2020-03-27 15:06:14 AirPods Apple Inc. 0x2002
6 2020-03-31 15:08:08 AirPods Pro Apple Inc. 0x200e
7 2020-04-13 15:26:44 iPhone Apple Inc. iPhone iPhone8,4 13.4.1
8 2020-04-13 21:16:00 Apple Watch Apple Inc. Watch Watch4,3 6.2.1
Execution finished without errors.
Result: 8 rows returned in 20ms
At line 1:
select datetime (creation_date+978307200, 'unixepoch','localtime') as timestamp, name, manufacturer, model, hardware, software from source_devices

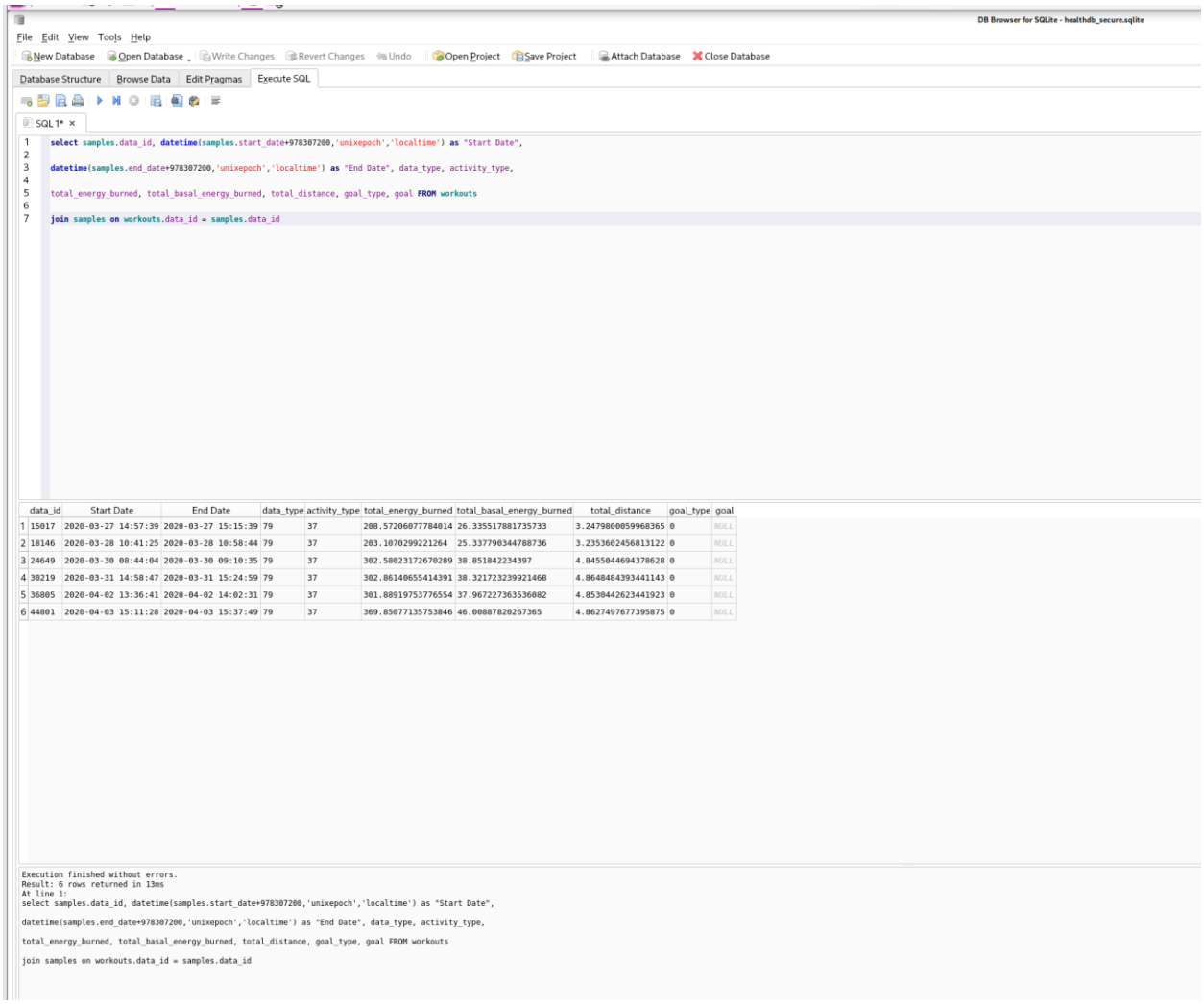

DB Browser for SQLite - healthdb.sqlite
File Edit View Tools Help
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database
Database Structure Browse Data Edit Pragmas Execute SQL
SQL 1*
1 select name, bundle_id, source_options, local_device, product_type from sources
2
name bundle_id source_options local_device product_type
1 This Is's iPhone com.apple.health.284D8298-B83A-422F-… 2 1 iPhone8,4
2 Health com.apple.Health 3 0
3 This Is's Apple Watch com.apple.health.8DDD238A-F8E1-41D2-… 2 0 Watch4,3
4 This Is's Apple Watch com.apple.health.9676B92F-913B-4B9D-… 2 0 Watch4,3
5 This Is's Apple Watch com.apple.health.9676B92F-913B-4B9D-… 2 0 Watch4,3
6 Sync Solver com.redshiftdev.Fitbit-Health-Sync 5 0
Execution finished without errors.
Result: 6 rows returned in 11ms
At line 1:
select name, bundle_id, source_options, local_device, product_type from sources

```
┌──(water㉿kalipurple)-[~/…/var/mobile/Library/Health]
└─$ sqlitebrowser healthdb_secure.sqlite
```





	Energy Burned Goal Date	cache_index	energy_burned	energy_burned_goal	active_hours	active_hours_goal	brisk_minutes	brisk_minutes_goal	steps	walk_distance
1	0000-12-30 05:00:00	2020-03-20 04:00:00	0.0	0.0	0.0	12.0	0.0	30.0	0.0	0.0
2	2020-03-22 04:51:53	2020-03-21 04:00:00	28.964999999999989	480.0	1.0	12.0	0.0	30.0	54.0	45.659500060081484
3	2020-03-22 04:51:53	2020-03-22 04:00:00	454.21100000000052	480.0	15.0	12.0	28.0	30.0	5288.0	4141.5635134530103
4	2020-03-22 04:51:53	2020-03-23 04:00:00	597.57099999999946	480.0	14.0	12.0	53.0	30.0	7930.0	6242.1125183487038
5	2020-03-22 04:51:53	2020-03-24 04:00:00	525.60100000000011	480.0	14.0	12.0	40.0	30.0	5951.0	4704.2290147113863
6	2020-03-22 04:51:53	2020-03-25 04:00:00	472.96200000000061	480.0	13.0	12.0	28.0	30.0	5560.0	4387.4990121531018
7	2020-03-22 04:51:53	2020-03-26 04:00:00	564.0249999999993	480.0	15.0	12.0	41.0	30.0	6814.0	5433.9835160946859
8	2020-03-22 04:51:53	2020-03-27 04:00:00	757.93500000000063	480.0	14.0	12.0	63.0	30.0	10094.0	8785.4152085254191
9	2020-03-22 04:51:53	2020-03-28 04:00:00	673.80099999999993	480.0	14.0	12.0	44.0	30.0	10367.0	9052.7562603347487
10	2020-03-22 04:51:53	2020-03-29 04:00:00	534.14900000000012	480.0	15.0	12.0	37.0	30.0	7762.0	6134.2684259701064
11	2020-03-22 04:51:53	2020-03-30 04:00:00	881.53400000000022	480.0	16.0	12.0	77.0	30.0	12580.0	11142.207718149897
12	2020-03-22 04:51:53	2020-03-31 04:00:00	841.444	480.0	16.0	12.0	67.0	30.0	11119.0	9976.2244686643535
13	2020-03-22 04:51:53	2020-04-01 04:00:00	576.73200000000043	480.0	15.0	12.0	44.0	30.0	7749.0	6229.0069276129489
14	2020-03-22 04:51:53	2020-04-03 04:00:00	753.33800000000042	480.0	11.0	12.0	62.0	30.0	8787.0	8299.9142082666626
15	2020-03-22 04:51:53	2020-04-02 04:00:00	812.37899999999956	480.0	14.0	12.0	65.0	30.0	10953.0	10099.51168533433
16	2020-03-22 04:51:53	2020-04-11 04:00:00	0.0	480.0	0.0	12.0	0.0	30.0	0.0	0.0
17	2020-03-22 04:51:53	2020-04-12 04:00:00	196.67	480.0	6.0	12.0	13.0	30.0	1633.0	1323.2071656180192
18	2020-03-22 04:51:53	2020-04-13 04:00:00	599.02899999999977	480.0	14.0	12.0	35.0	30.0	6642.0	5423.5833522011499
19	2020-03-22 04:51:53	2020-04-14 04:00:00	631.4399999999988	480.0	16.0	12.0	34.0	30.0	5970.0	4766.4376336293935
20	2020-03-22 04:51:53	2020-04-15 04:00:00	625.15600000000052	480.0	14.0	12.0	37.0	30.0	7442.0	6007.912840044497



routined Local.sqlite and locationd cache_encryptedB.db

```
┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ ls
audit     buddy           containers             empty     installd            Keychains   lock  'Managed Preferences'   MobileDevice           networkd      run      staged_system_apps   wireless
backups   cache           db                     folders   iomfb_bics_daemon   lib         log   mobile                  MobileSoftwareUpdate   preferences   select   tmp
binpack   checkra1n.dmg   dropbear_rsa_host_key  hardware  keybags             local       logs  MobileAsset             msgs                   root          spool    vm

┌──(water㉿kalipurple)-[~/…/Volumes/JOSH/NoTar-13-4-1/var]
└─$ cd root

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/root]
└─$ ls
Library  Media

┌──(water㉿kalipurple)-[~/…/JOSH/NoTar-13-4-1/var/root]
└─$ cd Library

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/root/Library]
└─$ ls
'Application Support'   Caches   Catacomb   com.apple.wifianalyticsd   Lockdown   Logs   MobileContainerManager   Preferences   USBRestricted

┌──(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/root/Library]
└─$ cd Caches

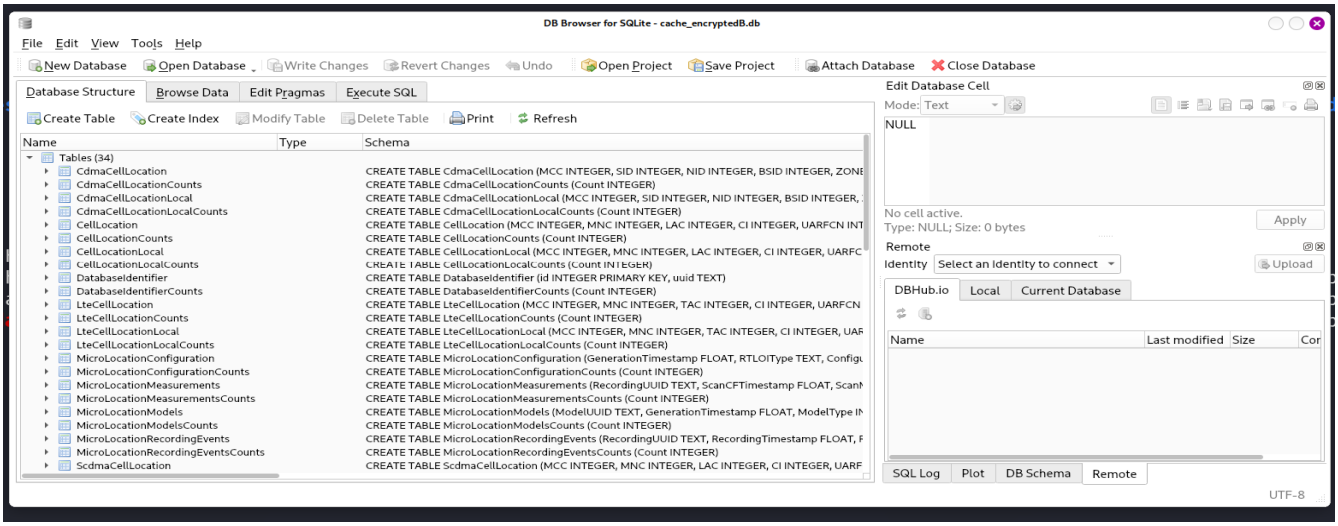
┌──(water㉿kalipurple)-[~/…/var/root/Library/Caches]
└─$ ls
Backup  com.apple.backupd  com.apple.containermanagerd  com.apple.coresymbolicationd  com.apple.mobileassetd  com.apple.pipelined  com.apple.transparencyd  com.apple.wifid  dscsym  hangtracerd  locationd

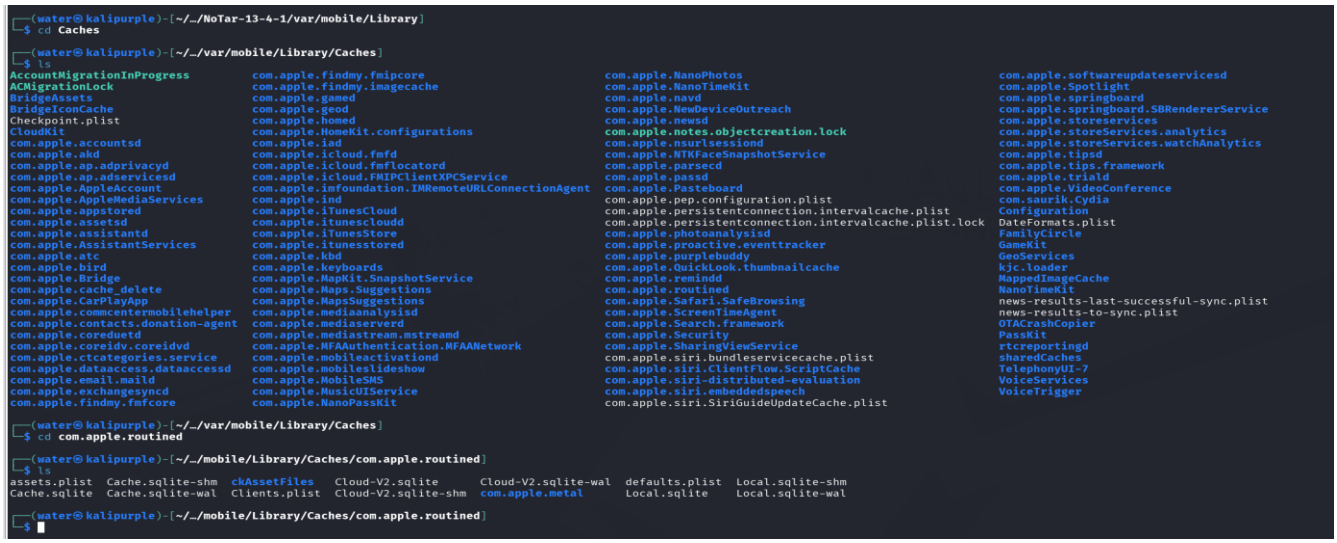
┌──(water㉿kalipurple)-[~/…/var/root/Library/Caches]
└─$ cd locationd

┌──(water㉿kalipurple)-[~/…/root/Library/Caches/locationd]
└─$ ls
adsf                                cache_encryptedB.db-shm  cache_encryptedC.db-wal  clients.plist    consolidated.db-shm  gyroCal.db-shm  locScoreInfo                    stats.plist   tiles_preload_manifest
bluetooth-carkit-whitelist.plist    cache_encryptedB.db-wal  cache.plist              clsef.bin        consolidated.db-wal  gyroCal.db-wal  sensorRecorder_encryptedC.db    tiles         tiles_transit_mac
bluetooth-carkit-whitelist-version.plist  cache_encryptedC.db  cldavl.avl           clsef.bin.gz     ephemeris            harvest         sensorRecorder_encryptedC.db-shm  tiles_app_v2  userinfo.plist
cache_encryptedB.db                 cache_encryptedC.db-shm  cldavl.avl.gz            consolidated.db  gyroCal.db           indoor_tiles    sensorRecorder_encryptedC.db-wal  tiles_cell    xtra2.bin

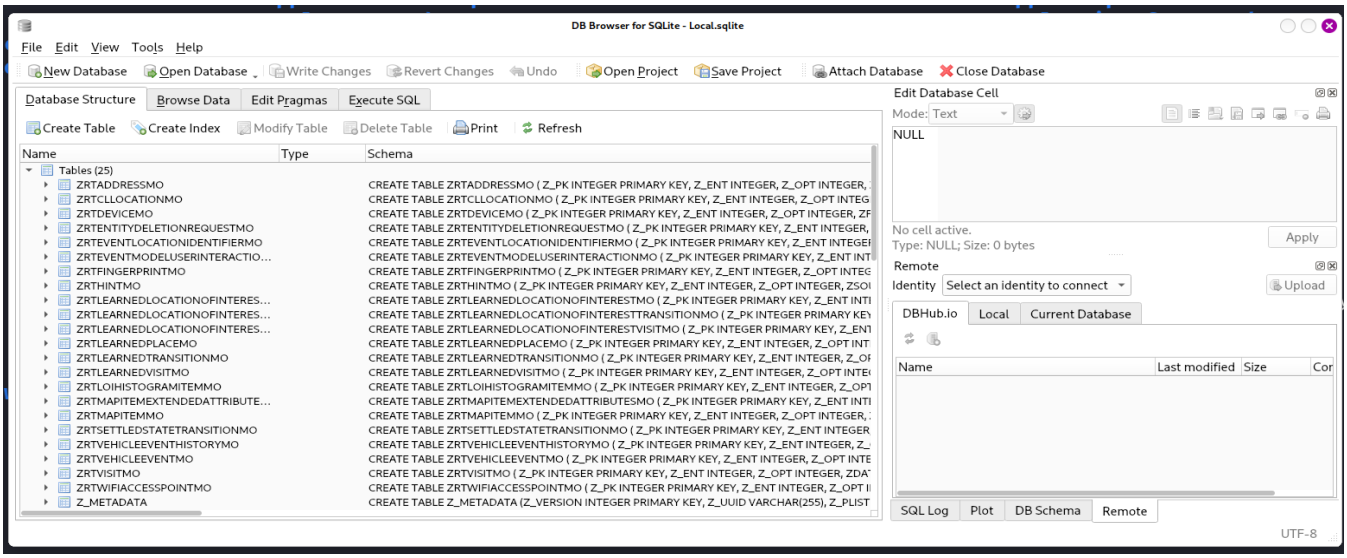
┌──(water㉿kalipurple)-[~/…/root/Library/Caches/locationd]
└─$
```

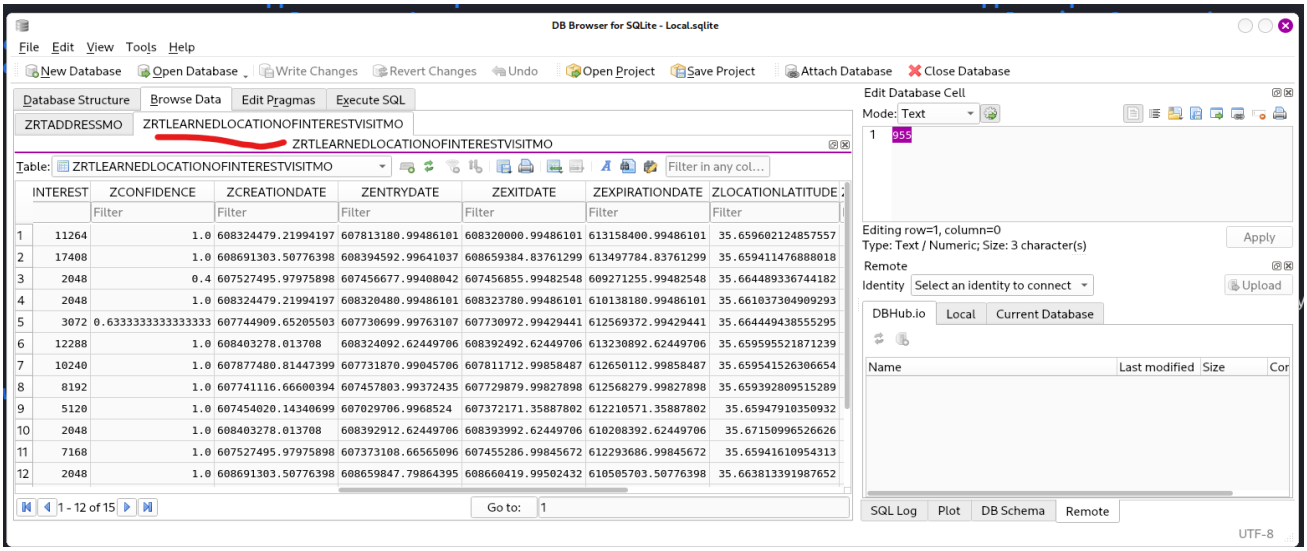
```
┌──(water㉿kalipurple)-[~/…/root/Library/Caches/locationd]
└─$ sqlitebrowser cache_encryptedB.db
```


DB Browser for SQLite - cache_encryptedB.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (34)
CdmaCellLocation
CdmaCellLocationCounts
CdmaCellLocationLocal
CdmaCellLocationLocalCounts
CellLocation
CellLocationCounts
CellLocationLocal
CellLocationLocalCounts
DatabaseIdentifier
DatabaseIdentifierCounts
LteCellLocation
LteCellLocationCounts
LteCellLocationLocal
LteCellLocationLocalCounts
MicroLocationConfiguration
MicroLocationConfigurationCounts
MicroLocationMeasurements
MicroLocationMeasurementsCounts
MicroLocationModels
MicroLocationModelsCounts
MicroLocationRecordingEvents
MicroLocationRecordingEventsCounts
ScdmaCellLocation
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8


(water㉿kalipurple)-[~/…/NoTar-13-4-1/var/mobile/Library]
$ cd Caches
(water㉿kalipurple)-[~/…/var/mobile/Library/Caches]
$ ls
AccountMigrationInProgress
ACMigrationLock
BridgeAssets
BridgeIconCache
Checkpoint.plist
CloudKit
com.apple.accountsd
com.apple.akd
com.apple.ap.adprivacyd
com.apple.ap.adservicesd
com.apple.AppleAccount
com.apple.AppleMediaServices
com.apple.appstored
com.apple.assetsd
com.apple.assistantd
com.apple.AssistantServices
com.apple.atc
com.apple.bird
com.apple.Bridge
com.apple.cache_delete
com.apple.CarPlayApp
com.apple.commcentermobilehelper
com.apple.contacts.donation-agent
com.apple.coreduetd
com.apple.coreidv.coreidvd
com.apple.ctcategories.service
com.apple.dataaccess.dataaccessd
com.apple.email.maild
com.apple.exchangesyncd
com.apple.findmy.fmfcore
com.apple.findmy.fmipcore
com.apple.findmy.imagecache
com.apple.gamed
com.apple.geod
com.apple.homed
com.apple.HomeKit.configurations
com.apple.iad
com.apple.icloud.fmfd
com.apple.icloud.fmflocatord
com.apple.icloud.FMIPClientXPCService
com.apple.imfoundation.IMRemoteURLConnectionAgent
com.apple.ind
com.apple.iTunesCloud
com.apple.itunescloudd
com.apple.iTunesStore
com.apple.itunesstored
com.apple.kbd
com.apple.keyboards
com.apple.MapKit.SnapshotService
com.apple.Maps.Suggestions
com.apple.MapsSuggestions
com.apple.mediaanalysisd
com.apple.mediaserverd
com.apple.mediastream.mstreamd
com.apple.MFAAuthentication.MFANetwork
com.apple.mobileactivationd
com.apple.mobileslideshow
com.apple.MobileSMS
com.apple.MusicUIService
com.apple.NanoPassKit
com.apple.NanoPhotos
com.apple.NanoTimeKit
com.apple.navd
com.apple.NewDeviceOutreach
com.apple.newsd
com.apple.notes.objectcreation.lock
com.apple.nsurlsessiond
com.apple.NTKFaceSnapshotService
com.apple.parsecd
com.apple.passd
com.apple.Pasteboard
com.apple.pep.configuration.plist
com.apple.persistentconnection.intervalcache.plist
com.apple.persistentconnection.intervalcache.plist.lock
com.apple.photoanalysisd
com.apple.proactive.eventtracker
com.apple.purplebuddy
com.apple.QuickLook.thumbnailcache
com.apple.remindd
com.apple.routined
com.apple.Safari.SafeBrowsing
com.apple.ScreenTimeAgent
com.apple.Search.framework
com.apple.Security
com.apple.SharingViewService
com.apple.siri.bundleservicecache.plist
com.apple.siri.ClientFlow.ScriptCache
com.apple.siri-distributed-evaluation
com.apple.siri.embeddedspeech
com.apple.siri.SiriGuideUpdateCache.plist
com.apple.softwareupdateservicesd
com.apple.Spotlight
com.apple.springboard
com.apple.springboard.SBRendererService
com.apple.storeservices
com.apple.storeServices.analytics
com.apple.storeServices.watchAnalytics
com.apple.tipsd
com.apple.tips.framework
com.apple.triald
com.apple.VideoConference
com.saurik.Cydia
Configuration
DateFormats.plist
FamilyCircle
GameKit
GeoServices
kjc.loader
MappedImageCache
NanoTimeKit
news-results-last-successful-sync.plist
news-results-to-sync.plist
OTACrashCopier
PassKit
rtcreportingd
sharedCaches
TelephonyUI-7
VoiceServices
VoiceTrigger
(water㉿kalipurple)-[~/…/var/mobile/Library/Caches]
$ cd com.apple.routined
(water㉿kalipurple)-[~/…/mobile/Library/Caches/com.apple.routined]
$ ls
assets.plist
Cache.sqlite
Cache.sqlite-shm
Cache.sqlite-wal
ckAssetFiles
Clients.plist
Cloud-V2.sqlite
Cloud-V2.sqlite-shm
Cloud-V2.sqlite-wal
com.apple.metal
defaults.plist
Local.sqlite
Local.sqlite-shm
Local.sqlite-wal
(water㉿kalipurple)-[~/…/mobile/Library/Caches/com.apple.routined]
$


(water㉿kalipurple)-[~/…/mobile/Library/Caches/com.apple.routined]
$ sqlitebrowser Local.sqlite


DB Browser for SQLite - Local.sqlite
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (25)
ZRTADDRESSMO
ZRTCLLOCATIONMO
ZRTDEVICEMO
ZRTENTITYDELETIONREQUESTMO
ZRTEVENTLOCATIONIDENTIFIERMO
ZRTEVENTMODELUSERINTERACTIO…
ZRTFINGERPRINTMO
ZRTHINTMO
ZRTLEARNEDLOCATIONOFINTERES…
ZRTLEARNEDLOCATIONOFINTERES…
ZRTLEARNEDLOCATIONOFINTERES…
ZRTLEARNEDPLACEMO
ZRTLEARNEDTRANSITIONMO
ZRTLEARNEDVISITMO
ZRTLOIHISTOGRAMITEMMO
ZRTMAPITEMEXTENDEDATTRIBUTE…
ZRTMAPITEMMO
ZRTSETTLEDSTATETRANSITIONMO
ZRTVEHICLEEVENTHISTORYMO
ZRTVEHICLEEVENTMO
ZRTVISITMO
ZRTWIFIACCESSPOINTMO
Z_METADATA
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8



Task 8 Timeline and conclusions

1. Create a timeline of at least 12 distinct events from four or more artefact families. Include raw time, field name, epoch and unit, UTC conversion, event description and evidence ID.

2. Present at least three correlations and explain two alternative interpretations. State what the evidence establishes, what remains uncertain and what further evidence would resolve the gaps.

No.	Raw Time	Field Name	Epoch and Unit	UTC Conversion	Event Description
1	607049916.117145	visit_time	Apple Epoch / Seconds	2020-03-28 00:58:36 UTC	Google search for MLB 2020 season.
2	607049917.431689	visit_time	Apple Epoch / Seconds	2020-03-28 00:58:37 UTC	Google search for when MLB starts in 2020.
3	607049943.212165	visit_time	Apple Epoch / Seconds	2020-03-28 00:59:03 UTC	Browsing activity relating to MLB 2020 season.
4	607049971.644170	visit_time	Apple Epoch / Seconds	2020-03-28 00:59:31 UTC	Another visit relating to MLB 2020 season.
5	607049972.930589	visit_time	Apple Epoch / Seconds	2020-03-28 00:59:32 UTC	Repeated browsing of MLB 2020 season information.
6	607050164.262689	visit_time	Apple Epoch / Seconds	2020-03-28 01:02:44 UTC	Google search concerning the NHL season resumption.

No.	Raw Time	Field Name	Epoch and Unit	UTC Conversion	Event Description
7	607050164.953563	visit_time	Apple Epoch / Seconds	2020-03-28 01:02:44 UTC	Another recorded visit to the NHL search results.
8	607050231.330998	visit_time	Apple Epoch / Seconds	2020-03-28 01:03:51 UTC	Browsing an article about the NHL resuming its season.
9	607050233.893370	visit_time	Apple Epoch / Seconds	2020-03-28 01:03:53 UTC	Another recorded visit concerning the NHL season.
10	607050461.869823	visit_time	Apple Epoch / Seconds	2020-03-28 01:07:41 UTC	Browsing the Apple iPad Pro webpage.
11	607052629.819015	visit_time	Apple Epoch / Seconds	2020-03-28 01:43:49 UTC	Browsing a webpage titled DFIR Review.
12	607052677.913083	visit_time	Apple Epoch / Seconds	2020-03-28 01:44:37 UTC	Browsing a webpage titled Mobile Installation Logs - DFIR.

Correlation 1: Safari Browsing and Search Activities

The Safari browsing records show searches relating to the MLB and NHL seasons within a short period on 28 March 2020.

This suggests that the device recorded multiple browsing activities relating to sports information.

Alternative interpretations:

- The same user may have been researching sports news.
- The browsing records may have resulted from repeated searches, page reloads or automated browser activity.

Correlation 2: Messages and Contact Records

The SMS database and AddressBook database provide opportunities to compare message participants with saved contact information.

A matching phone number between a message record and a stored contact can help establish an association between the records.

Alternative interpretations:

- The phone number may belong to a saved contact.
- The number may have been stored under an outdated or incorrect contact name, or the message may have originated from another person using that number.

Correlation 3: Photographs and KnowledgeC Activity

The Photos database contains media records, including filenames and asset information, while KnowledgeC contains records relating to application and device activities.

Comparing their timestamps may help establish whether photo-related activity occurred around the same period as application activity.

Alternative interpretations:

- The photograph may have been captured directly using the iPhone.
- The photograph may have been imported, downloaded or received from another source.

Evidence Establishes

The examination establishes that:

1. The supplied evidence relates to an iPhone SE running iOS 13.4.1.
2. The device contains Safari browsing history with recorded activities dated 28 March 2020.
3. The Photos database contains media records associated with image files.
4. The KnowledgeC database contains structured records relating to device and application activities.
5. The device also contains SMS, contact, calendar, health and location-related forensic artefacts.

What Remains Uncertain

Some issues cannot be conclusively established from the available screenshots:

1. The identity of the individual physically operating the device.
2. Whether the device owner personally performed every recorded activity.
3. Whether photographs were captured by the device or received from another source.
4. Whether recorded location information represents the actual physical location of the device at a particular time.
5. Whether browsing and messaging activities were performed by the same individual.

2.7. Analysis and Findings

The investigation examined an iPhone SE running iOS 13.4.1 using different forensic databases and artefacts. Safari browsing history revealed recorded internet activities relating to sports information and other websites. SMS, contacts, photographs, KnowledgeC and location records were also examined to identify relevant device activities. The recovered timestamps help establish relationships between browsing, communication and other activities

2.8. Challenges and Solutions

One of the major challenges was understanding the different iOS forensic databases and their structures. Converting Apple timestamps into readable UTC dates also required additional attention. Some records were difficult to interpret because of the different timestamp formats and database fields.

2.9. Conclusion

The investigation provided practical experience in examining and analysing digital evidence from an iPhone device. Different artefacts, including browsing history, messages, photographs and application records, helped provide information about device activities. The examination also demonstrated how timestamps can be used to reconstruct a timeline of events.

2.10. Recommendations

Assignment like this improved student knowledge and understanding on Forensic case giving the student the ability to think deep and do deep research before concluding on a case

2.11. References

Class materials Guide from Founder ICDFA on Basic Networking Skills for Digital Forensics, (MobileIoT Forensic Case Study)

https://onedrive.live.com/?id=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments%2FDocuments%2FDigital%20Forensics%2FMobileIoT%20Forensics%20Case%20Study%2FiOS&listurl=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments&redeem=aHR0cHM6Ly8xZHJ2Lm1zL2YvYy9hNjc0OTA4YTVlMmUyZjJhL0lnQ3cyMHc1dFFkUFRKX21rTy05clRYUUFSWTNJVU9jWVExNTNyRkU1bUdIZWtzP2U9RGt1YkRM